



American International University-Bangladesh (AIUB)

**Department of Computer Science and Engineering (CSE)
Faculty of Science & Technology (FST)
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Section: E

**Mid Term Project: Handling Missing Values using R for this
given “Maternal Health and Caesarian Section Outcomes
dataset”**

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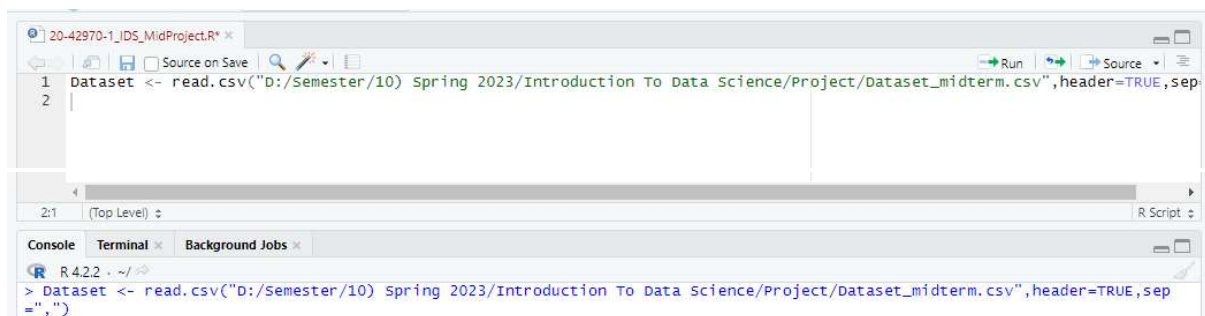
Submitted Date: 13th March 2023

Handling Missing Values using R for this given “Maternal Health and Caesarian Section Outcomes” Dataset

Overview: The "Maternal Health and Caesarian Section Outcomes" dataset is a curated collection of 80 items that discuss numerous maternal and health-related variables that may impact the choice to have a caesarean section. It takes into account both demographic and health-related factors such as the mother's age, weight, number of previous births, delivery time, blood pressure status, existence of cardiac issues, and if the delivery was caesarian. This dataset is useful to health researchers and practitioners, since it provides a solid foundation for examining the complicated interplay between maternal health characteristics and caesarean section outcomes. The information provides insights into how age and weight, as well as certain health issues, link with the chance of a caesarean birth. It is intended not only to improve understanding of the factors that influence caesarean sections, but also to aid in the creation of targeted treatments and policies to improve maternal and newborn health outcomes.

1) Import the Data Set

```
Dataset <- read.csv("D:/Semester/10) Spring 2023/Introduction To Data Science/Project/Dataset_midterm.csv",header=TRUE,sep=",")
```



The screenshot shows the R Studio interface. The script editor contains the following code:

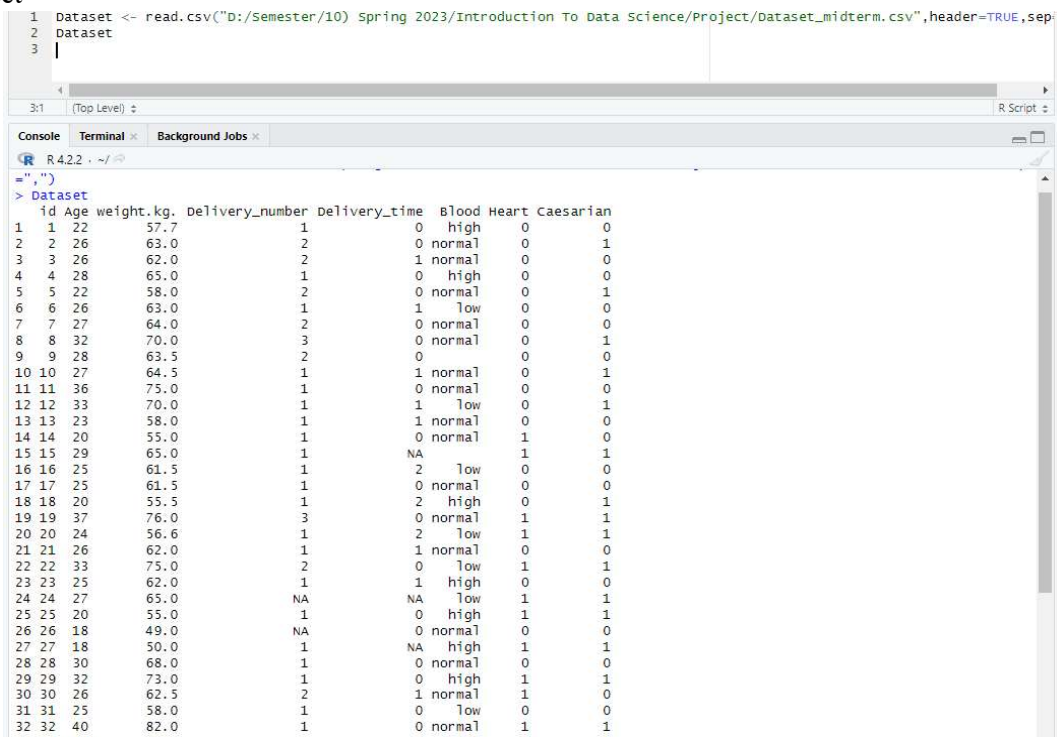
```
1 Dataset <- read.csv("D:/Semester/10) Spring 2023/Introduction To Data Science/Project/Dataset_midterm.csv",header=TRUE,sep=",")
2
```

The console shows the execution of the code:

```
> Dataset <- read.csv("D:/Semester/10) Spring 2023/Introduction To Data Science/Project/Dataset_midterm.csv",header=TRUE,sep=",")
```

2) Print the Data Set

Dataset



The screenshot shows the R Studio console with the command `Dataset` entered, which prints the first 32 rows of the dataset. The data is as follows:

	id	Age	weight.kg.	delivery_number	delivery_time	Blood	Heart	Caesarian
1	1	22	57.7	1	0	high	0	0
2	2	26	63.0	2	0	normal	0	1
3	3	26	62.0	2	1	normal	0	0
4	4	28	65.0	1	0	high	0	0
5	5	22	58.0	2	0	normal	0	1
6	6	26	63.0	1	1	low	0	0
7	7	27	64.0	2	0	normal	0	0
8	8	32	70.0	3	0	normal	0	1
9	9	28	63.5	2	0	0	0	0
10	10	27	64.5	1	1	normal	0	1
11	11	36	75.0	1	0	normal	0	0
12	12	33	70.0	1	1	low	0	1
13	13	23	58.0	1	1	normal	0	0
14	14	20	55.0	1	0	normal	1	0
15	15	29	65.0	1	NA	1	1	1
16	16	25	61.5	1	2	low	0	0
17	17	25	61.5	1	0	normal	0	0
18	18	20	55.5	1	2	high	0	1
19	19	37	76.0	3	0	normal	1	1
20	20	24	56.6	1	2	low	1	1
21	21	26	62.0	1	1	normal	0	0
22	22	33	75.0	2	0	low	1	1
23	23	25	62.0	1	1	high	0	0
24	24	27	65.0	NA	NA	low	1	1
25	25	20	55.0	1	0	high	1	1
26	26	18	49.0	NA	0	normal	0	0
27	27	18	50.0	1	NA	high	1	1
28	28	30	68.0	1	0	normal	0	0
29	29	32	73.0	1	0	high	1	1
30	30	26	62.5	2	1	normal	1	0
31	31	25	58.0	1	0	low	0	0
32	32	40	82.0	1	0	normal	1	1

3) Show The Attributes Name from the Data Set

names(Dataset)

```
20-42970-1_IDS_MidProject.R
1 Dataset <- read.csv("D:/Semester/10) Spring 2023/Introduction To Data Science/Project/Dataset_midterm.csv",header=TRUE,sep
2 Dataset
3 names(Dataset)

> names(Dataset)
[1] "id"          "Age"          "weight.kg."   "delivery_number" "delivery_time" "Blood"
[7] "Heart"       "caesarian"
```

4) Summary of every attribute from the Data Set

summary(Dataset)

```
20-42970-1_IDS_MidProject.R
1 Dataset <- read.csv("D:/Semester/10) Spring 2023/Introduction To Data Science/Project/Dataset_midterm.csv",header=TRUE,sep
2 Dataset
3 names(Dataset)
4 summary(Dataset)

4:1 (Top Level)
R Script

Console Terminal Background Jobs
R 4.2.2 ~
> summary(Dataset)
      id      Age      weight.kg.  delivery_number delivery_time      Blood      Heart
Min.   :1.00   Min.   :18.00   Min.   : 49.00   Min.   :1.000   Min.   :0.0000   Length:80   Min.   :0.000
1st Qu.:20.75  1st Qu.:25.00   1st Qu.: 61.00   1st Qu.:1.000   1st Qu.:0.0000   class :character 1st Qu.:0.000
Median :40.50  Median :28.00   Median : 63.50   Median :1.500   Median :0.0000   Mode  :character   Median :0.000
Mean   :40.50  Mean   :29.68   Mean   : 65.13   Mean   :1.679   Mean   :0.6234                Mean :0.375
3rd Qu.:60.25  3rd Qu.:32.00   3rd Qu.: 68.00   3rd Qu.:2.000   3rd Qu.:1.0000              3rd Qu.:1.000
Max.   :80.00  Max.   :95.00   Max.   :110.00   Max.   :4.000   Max.   :2.0000              Max.   :1.000

      Caesarian
Min.   :0.0000
1st Qu.:0.0000
Median :1.0000
Mean   :0.5641
3rd Qu.:1.0000
Max.   :1.0000
NA's   :2
```

5) Find all types of data from a Data Set

str(Dataset)

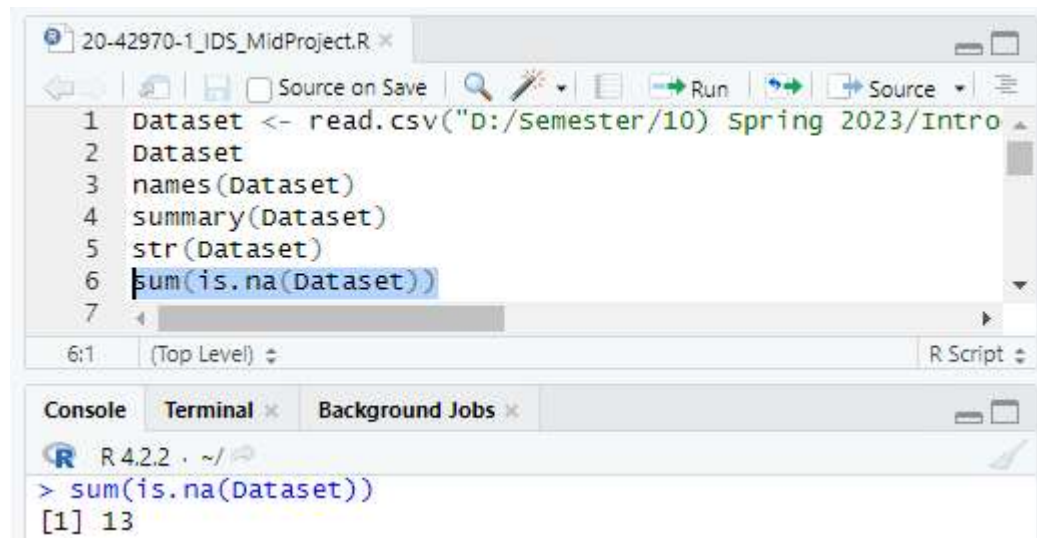
```
20-42970-1_IDS_MidProject.R
1 Dataset <- read.csv("D:/Semester/10) Spring 2023/Introduction To Data Science/Project/Dataset_midterm.csv",header=TRUE,sep
2 Dataset
3 names(Dataset)
4 summary(Dataset)
5 str(Dataset)

5:1 (Top Level)
R Script

Console Terminal Background Jobs
R 4.2.2 ~
      Caesarian      NA's :3      NA's :3      NA's :2      NA's :3
Min.   :0.0000
1st Qu.:0.0000
Median :1.0000
Mean   :0.5641
3rd Qu.:1.0000
Max.   :1.0000
NA's   :2
> str(Dataset)
'data.frame': 80 obs. of 8 variables:
 $ id      : int  1 2 3 4 5 6 7 8 9 10 ...
 $ Age     : int  22 26 26 28 22 26 27 32 28 27 ...
 $ weight.kg.: num  57.7 63 62 65 58 63 64 70 63.5 64.5 ...
 $ delivery_number: int  1 2 2 1 2 1 2 3 2 1 ...
 $ delivery_time : int  0 0 1 0 0 1 0 0 0 1 ...
 $ Blood     : chr  "high" "normal" "normal" "high" ...
 $ Heart     : int  0 0 0 0 0 0 0 0 0 ...
 $ Caesarian  : int  0 1 0 0 1 0 0 1 0 1 ...
```

6) Show the Sum of The Data Set

`sum(is.na(Dataset))`



The screenshot shows the R Studio interface. The script editor contains the following code:

```
1 Dataset <- read.csv("D:/Semester/10) Spring 2023/Intro
2 Dataset
3 names(Dataset)
4 summary(Dataset)
5 str(Dataset)
6 sum(is.na(Dataset))
7
```

The console output shows the result of the command:

```
> sum(is.na(Dataset))
[1] 13
```

7) Show the Types of Data for All Attributes.

`mode(Dataset$id)`

`mode(Dataset$Age)`

`mode(Dataset$weight.kg.)`

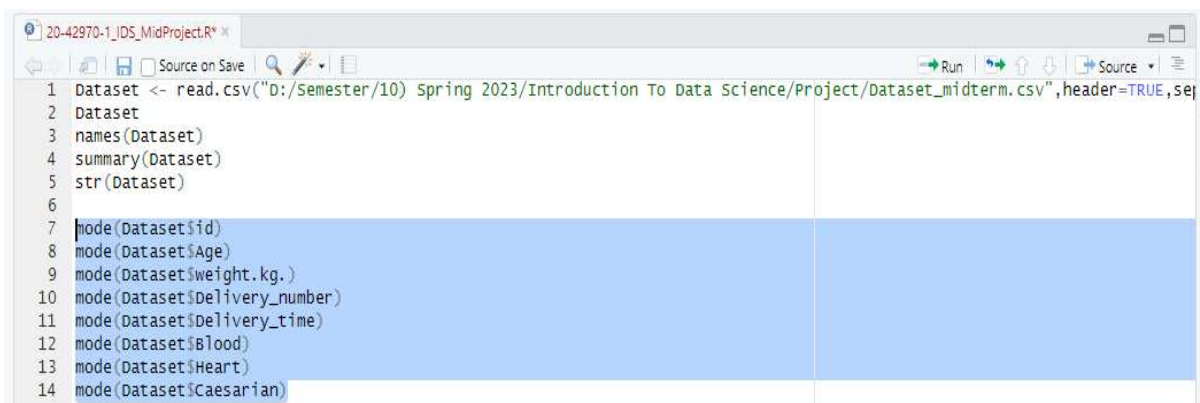
`mode(Dataset$Delivery_number)`

`mode(Dataset$Delivery_time)`

`mode(Dataset$Blood)`

`mode(Dataset$Heart)`

`mode(Dataset$Caesarian)`



The screenshot shows the R Studio interface. The script editor contains the following code:

```
1 Dataset <- read.csv("D:/Semester/10) Spring 2023/Introduction To Data Science/Project/Dataset_midterm.csv",header=TRUE,sep
2 Dataset
3 names(Dataset)
4 summary(Dataset)
5 str(Dataset)
6
7 mode(Dataset$id)
8 mode(Dataset$Age)
9 mode(Dataset$weight.kg.)
10 mode(Dataset$Delivery_number)
11 mode(Dataset$Delivery_time)
12 mode(Dataset$Blood)
13 mode(Dataset$Heart)
14 mode(Dataset$Caesarian)
```

```
R 4.2.2 . ~/
7:1 (Top Level)
R Script

Console Terminal Background Jobs

$ Blood      : chr "high" "normal" "normal" "high" ...
$ Heart      : int  0 0 0 0 0 0 0 0 0 ...
$ Caesarian  : int  0 1 0 0 1 0 0 1 0 1 ...

> mode(Dataset$id)
[1] "numeric"
> mode(Dataset$Age)
[1] "numeric"
> mode(Dataset$weight.kg.)
[1] "numeric"
> mode(Dataset$Delivery_number)
[1] "numeric"
> mode(Dataset$Delivery_time)
[1] "numeric"
> mode(Dataset$Blood)
[1] "character"
> mode(Dataset$Heart)
[1] "numeric"
> mode(Dataset$Caesarian)
[1] "numeric"
```

8) Compute the mean and median value of the Heart

```
meanHeart = mean(Dataset$Heart)
```

```
print(meanHeart)
```

```
medianHeart = median(Dataset$Heart)
```

```
print(medianHeart)
```

```
20-42970-1_IDS_MidProject.R*
Source on Save

1 Dataset <- read.csv("D:/Semester/10")
2 Dataset
3 names(Dataset)
4 summary(Dataset)
5 str(Dataset)
6
7 mode(Dataset$id)
8 mode(Dataset$Age)
9 mode(Dataset$weight.kg.)
10 mode(Dataset$Delivery_number)
11 mode(Dataset$Delivery_time)
12 mode(Dataset$Blood)
13 mode(Dataset$Heart)
14 mode(Dataset$Caesarian)
15
16 meanHeart = mean(Dataset$Heart)
17 print(meanHeart)
18
19 medianHeart = median(Dataset$Heart)
20 print(medianHeart)
```

```
19:1 (Top Level)
R 4.2.2 . ~/

> meanHeart = mean(Dataset$Heart)
> print(meanHeart)
[1] 0.375
> medianHeart = median(Dataset$Heart)
> print(medianHeart)
[1] 0
```



```
names(sort(table(Dataset$weight.kg)))
```

```
names(sort(table(Dataset$Blood)))
```

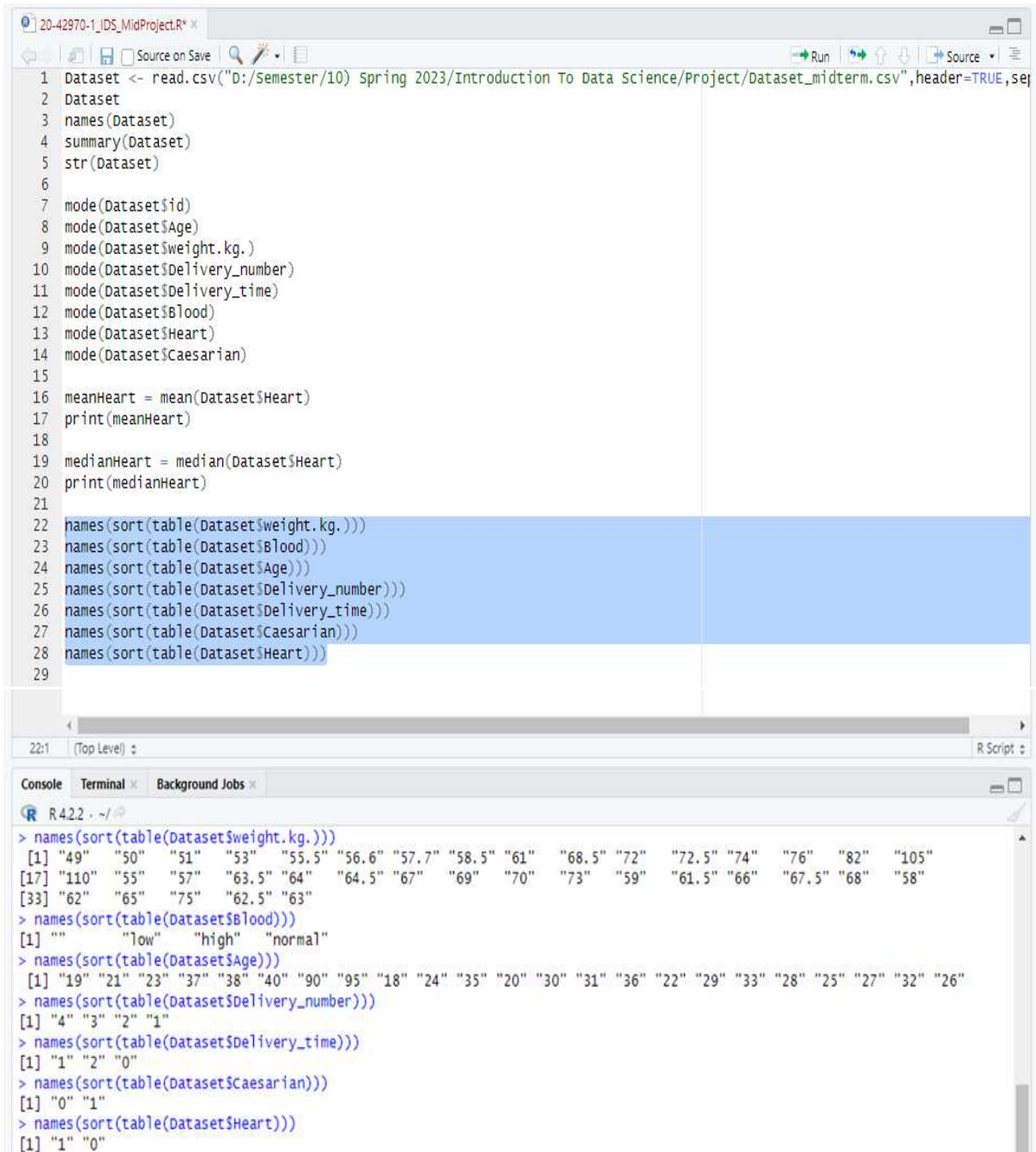
```
names(sort(table(Dataset$Age)))
```

```
names(sort(table(Dataset$Delivery_number)))
```

```
names(sort(table(Dataset$Delivery_time)))
```

```
names(sort(table(Dataset$Caesarian)))
```

```
names(sort(table(Dataset$Heart)))
```



The screenshot shows an R Studio window with a script editor and a console. The script editor contains R code for loading a CSV file, summarizing it, and calculating various statistics. The console shows the output of these commands.

```
20-42970-1_IDS_MidProject.R
Source on Save
Run
Source

1 Dataset <- read.csv("D:/Semester/10) Spring 2023/Introduction To Data Science/Project/Dataset_midterm.csv",header=TRUE,sep
2 Dataset
3 names(Dataset)
4 summary(Dataset)
5 str(Dataset)
6
7 mode(Dataset$id)
8 mode(Dataset$Age)
9 mode(Dataset$weight.kg.)
10 mode(Dataset$Delivery_number)
11 mode(Dataset$Delivery_time)
12 mode(Dataset$Blood)
13 mode(Dataset$Heart)
14 mode(Dataset$Caesarian)
15
16 meanHeart = mean(Dataset$Heart)
17 print(meanHeart)
18
19 medianHeart = median(Dataset$Heart)
20 print(medianHeart)
21
22 names(sort(table(Dataset$weight.kg.)))
23 names(sort(table(Dataset$Blood)))
24 names(sort(table(Dataset$Age)))
25 names(sort(table(Dataset$Delivery_number)))
26 names(sort(table(Dataset$Delivery_time)))
27 names(sort(table(Dataset$Caesarian)))
28 names(sort(table(Dataset$Heart)))
29
```

Console Output:

```
R 4.2.2 ~
> names(sort(table(Dataset$weight.kg.)))
[1] "49" "50" "51" "53" "55.5" "56.6" "57.7" "58.5" "61" "68.5" "72" "72.5" "74" "76" "82" "105"
[17] "110" "55" "57" "63.5" "64" "64.5" "67" "69" "70" "73" "59" "61.5" "66" "67.5" "68" "58"
[33] "62" "65" "75" "62.5" "63"
> names(sort(table(Dataset$Blood)))
[1] "" "low" "high" "normal"
> names(sort(table(Dataset$Age)))
[1] "19" "21" "23" "37" "38" "40" "90" "95" "18" "24" "35" "20" "30" "31" "36" "22" "29" "33" "28" "25" "27" "32" "26"
> names(sort(table(Dataset$Delivery_number)))
[1] "4" "3" "2" "1"
> names(sort(table(Dataset$Delivery_time)))
[1] "1" "2" "0"
> names(sort(table(Dataset$Caesarian)))
[1] "0" "1"
> names(sort(table(Dataset$Heart)))
[1] "1" "0"
```

9) Show the Standard Deviation of all attributes.

```
sdweight= sd(Dataset$weight.kg.)
print(sdweight)

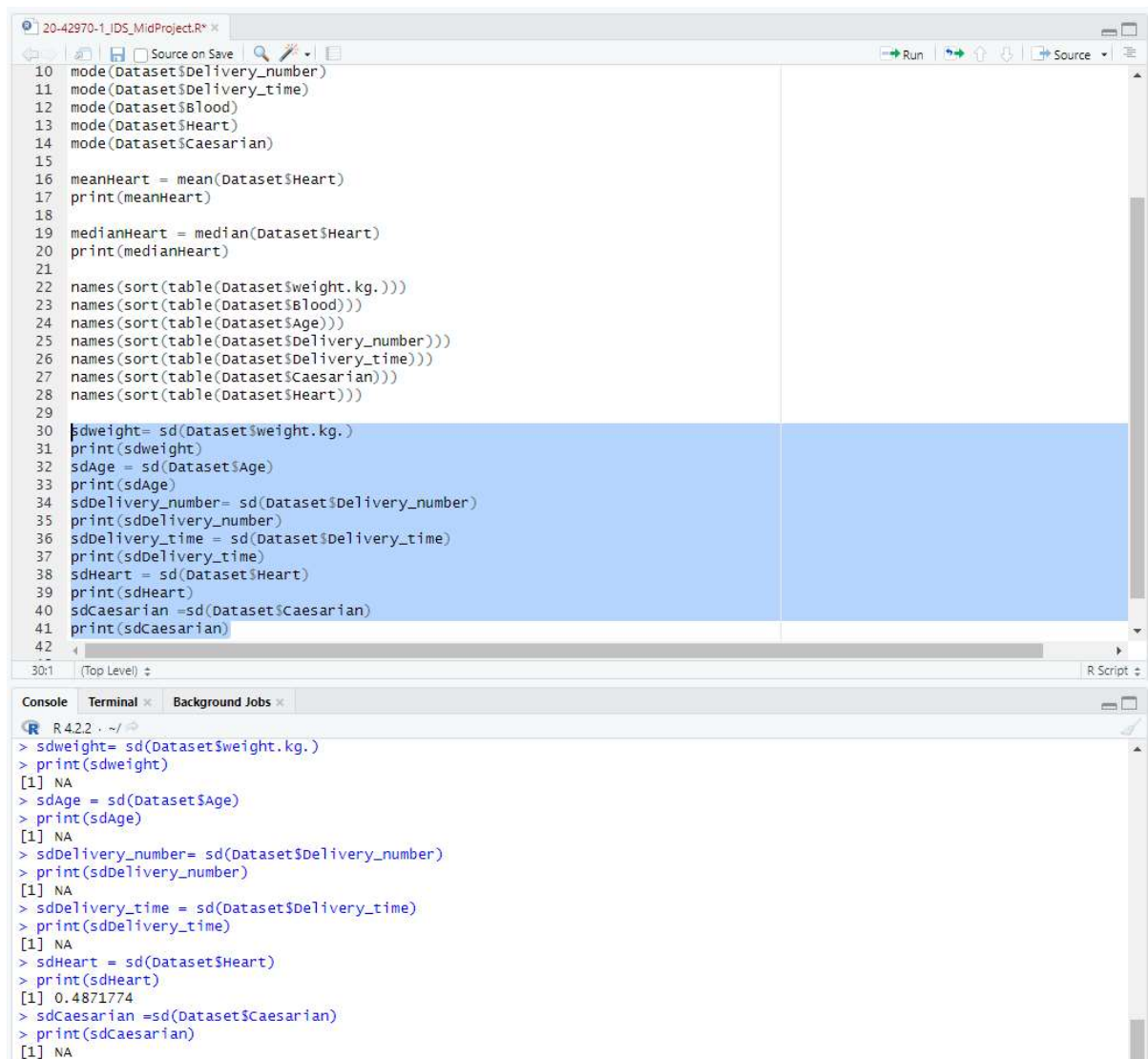
sdAge = sd(Dataset$Age)
print(sdAge)

sdDelivery_number= sd(Dataset$Delivery_number)
print(sdDelivery_number)

sdDelivery_time = sd(Dataset$Delivery_time)
print(sdDelivery_time)

sdHeart = sd(Dataset$Heart)
print(sdHeart)

sdCaesarian =sd(Dataset$Caesarian)
print(sdCaesarian)
```



The screenshot displays the R Studio environment. The script editor on the left contains R code for calculating standard deviations for various attributes of a dataset. The console at the bottom shows the execution of this code, with most standard deviation calculations resulting in NA, except for 'sdHeart' which returns 0.4871774.

```
10 mode(Dataset$Delivery_number)
11 mode(Dataset$Delivery_time)
12 mode(Dataset$Blood)
13 mode(Dataset$Heart)
14 mode(Dataset$Caesarian)
15
16 meanHeart = mean(Dataset$Heart)
17 print(meanHeart)
18
19 medianHeart = median(Dataset$Heart)
20 print(medianHeart)
21
22 names(sort(table(Dataset$weight.kg.)))
23 names(sort(table(Dataset$Blood)))
24 names(sort(table(Dataset$Age)))
25 names(sort(table(Dataset$Delivery_number)))
26 names(sort(table(Dataset$Delivery_time)))
27 names(sort(table(Dataset$Caesarian)))
28 names(sort(table(Dataset$Heart)))
29
30 sdweight= sd(Dataset$weight.kg.)
31 print(sdweight)
32 sdAge = sd(Dataset$Age)
33 print(sdAge)
34 sdDelivery_number= sd(Dataset$Delivery_number)
35 print(sdDelivery_number)
36 sdDelivery_time = sd(Dataset$Delivery_time)
37 print(sdDelivery_time)
38 sdHeart = sd(Dataset$Heart)
39 print(sdHeart)
40 sdCaesarian =sd(Dataset$Caesarian)
41 print(sdCaesarian)
42
```

Console Output:

```
> sdweight= sd(Dataset$weight.kg.)
> print(sdweight)
[1] NA
> sdAge = sd(Dataset$Age)
> print(sdAge)
[1] NA
> sdDelivery_number= sd(Dataset$Delivery_number)
> print(sdDelivery_number)
[1] NA
> sdDelivery_time = sd(Dataset$Delivery_time)
> print(sdDelivery_time)
[1] NA
> sdHeart = sd(Dataset$Heart)
> print(sdHeart)
[1] 0.4871774
> sdCaesarian =sd(Dataset$Caesarian)
> print(sdCaesarian)
[1] NA
```

10) Show the range of all attributes.

```
idRange=max(Dataset$id, na.rm=TRUE)-min(Dataset$id, na.rm=TRUE)

print(idRange)

AgeRange=max(Dataset$Age, na.rm=TRUE)-min(Dataset$Age, na.rm=TRUE)

print(AgeRange)

weightRange=max(Dataset$weight.kg., na.rm=TRUE)-min(Dataset$weight.kg., na.rm=TRUE)

print(weightRange)

Delivery_numberRange=max(Dataset$Delivery_number, na.rm=TRUE)-
min(Dataset$Delivery_number, na.rm=TRUE)

print(Delivery_numberRange)

Delivery_timeRange=max(Dataset$Delivery_time, na.rm=TRUE)-min(Dataset$Delivery_time,
na.rm=TRUE)

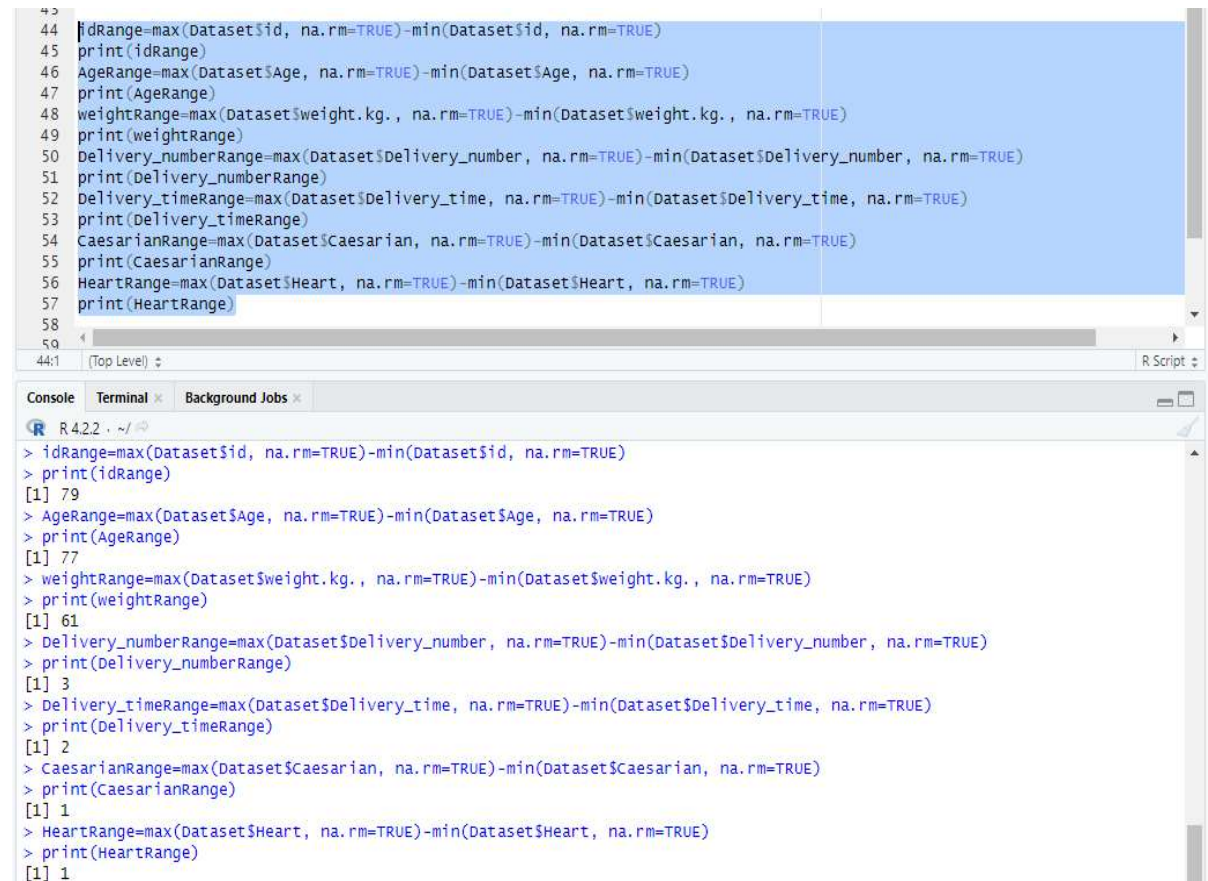
print(Delivery_timeRange)

CaesarianRange=max(Dataset$Caesarian, na.rm=TRUE)-min(Dataset$Caesarian, na.rm=TRUE)

print(CaesarianRange)

HeartRange=max(Dataset$Heart, na.rm=TRUE)-min(Dataset$Heart, na.rm=TRUE)

print(HeartRange)
```



The screenshot displays the R Studio interface. The top pane shows a script editor with the following R code:

```
44 idRange=max(Dataset$id, na.rm=TRUE)-min(Dataset$id, na.rm=TRUE)
45 print(idRange)
46 AgeRange=max(Dataset$Age, na.rm=TRUE)-min(Dataset$Age, na.rm=TRUE)
47 print(AgeRange)
48 weightRange=max(Dataset$weight.kg., na.rm=TRUE)-min(Dataset$weight.kg., na.rm=TRUE)
49 print(weightRange)
50 Delivery_numberRange=max(Dataset$Delivery_number, na.rm=TRUE)-min(Dataset$Delivery_number, na.rm=TRUE)
51 print(Delivery_numberRange)
52 Delivery_timeRange=max(Dataset$Delivery_time, na.rm=TRUE)-min(Dataset$Delivery_time, na.rm=TRUE)
53 print(Delivery_timeRange)
54 CaesarianRange=max(Dataset$Caesarian, na.rm=TRUE)-min(Dataset$Caesarian, na.rm=TRUE)
55 print(CaesarianRange)
56 HeartRange=max(Dataset$Heart, na.rm=TRUE)-min(Dataset$Heart, na.rm=TRUE)
57 print(HeartRange)
58
59
```

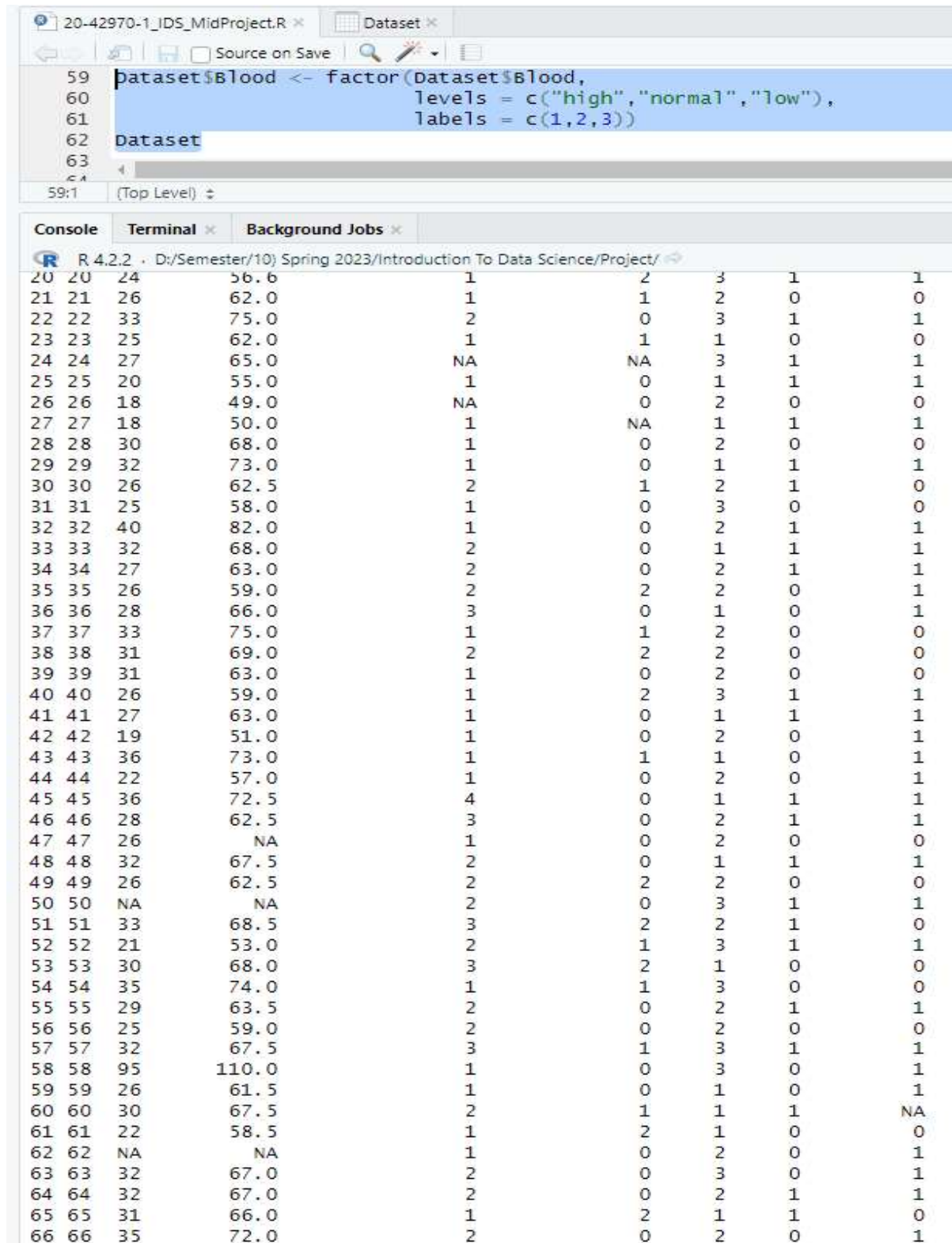
The bottom pane shows the R console output for the same code:

```
> idRange=max(Dataset$id, na.rm=TRUE)-min(Dataset$id, na.rm=TRUE)
> print(idRange)
[1] 79
> AgeRange=max(Dataset$Age, na.rm=TRUE)-min(Dataset$Age, na.rm=TRUE)
> print(AgeRange)
[1] 77
> weightRange=max(Dataset$weight.kg., na.rm=TRUE)-min(Dataset$weight.kg., na.rm=TRUE)
> print(weightRange)
[1] 61
> Delivery_numberRange=max(Dataset$Delivery_number, na.rm=TRUE)-min(Dataset$Delivery_number, na.rm=TRUE)
> print(Delivery_numberRange)
[1] 3
> Delivery_timeRange=max(Dataset$Delivery_time, na.rm=TRUE)-min(Dataset$Delivery_time, na.rm=TRUE)
> print(Delivery_timeRange)
[1] 2
> CaesarianRange=max(Dataset$Caesarian, na.rm=TRUE)-min(Dataset$Caesarian, na.rm=TRUE)
> print(CaesarianRange)
[1] 1
> HeartRange=max(Dataset$Heart, na.rm=TRUE)-min(Dataset$Heart, na.rm=TRUE)
> print(HeartRange)
[1] 1
```


11) Annotate as high as 1, as normal as 2, and as low as 3 from the Blood attribute.

```
Dataset$Blood <- factor(Dataset$Blood,
  levels = c("high","normal","low"),
  labels = c(1,2,3))
```

Dataset



The screenshot shows an R Studio window with a script editor and a console. The script editor contains the following code:

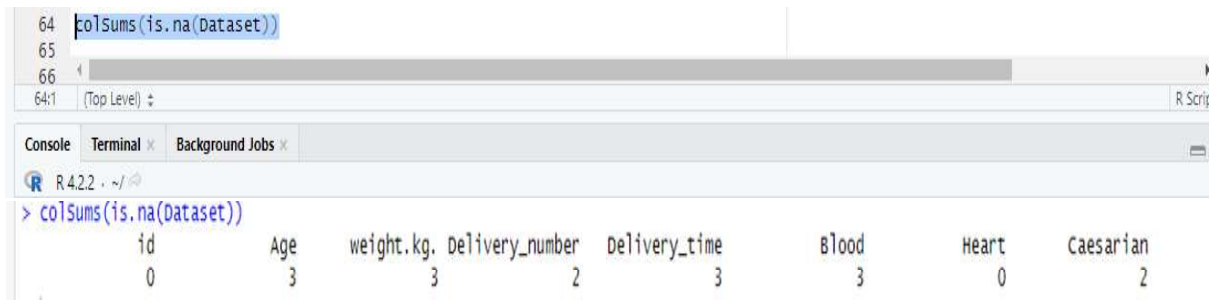
```
59 Dataset$Blood <- factor(Dataset$Blood,
60   levels = c("high","normal","low"),
61   labels = c(1,2,3))
62 Dataset
```

The console shows the output of the script, which is a data frame with 66 rows and 8 columns. The columns are labeled 1 through 8. The data is as follows:

	1	2	3	4	5	6	7	8
20	20	24	56.6	1	2	3	1	1
21	21	26	62.0	1	1	2	0	0
22	22	33	75.0	2	0	3	1	1
23	23	25	62.0	1	1	1	0	0
24	24	27	65.0	NA	NA	3	1	1
25	25	20	55.0	1	0	1	1	1
26	26	18	49.0	NA	0	2	0	0
27	27	18	50.0	1	NA	1	1	1
28	28	30	68.0	1	0	2	0	0
29	29	32	73.0	1	0	1	1	1
30	30	26	62.5	2	1	2	1	0
31	31	25	58.0	1	0	3	0	0
32	32	40	82.0	1	0	2	1	1
33	33	32	68.0	2	0	1	1	1
34	34	27	63.0	2	0	2	1	1
35	35	26	59.0	2	2	2	0	1
36	36	28	66.0	3	0	1	0	1
37	37	33	75.0	1	1	2	0	0
38	38	31	69.0	2	2	2	0	0
39	39	31	63.0	1	0	2	0	0
40	40	26	59.0	1	2	3	1	1
41	41	27	63.0	1	0	1	1	1
42	42	19	51.0	1	0	2	0	1
43	43	36	73.0	1	1	1	0	1
44	44	22	57.0	1	0	2	0	1
45	45	36	72.5	4	0	1	1	1
46	46	28	62.5	3	0	2	1	1
47	47	26	NA	1	0	2	0	0
48	48	32	67.5	2	0	1	1	1
49	49	26	62.5	2	2	2	0	0
50	50	NA	NA	2	0	3	1	1
51	51	33	68.5	3	2	2	1	0
52	52	21	53.0	2	1	3	1	1
53	53	30	68.0	3	2	1	0	0
54	54	35	74.0	1	1	3	0	0
55	55	29	63.5	2	0	2	1	1
56	56	25	59.0	2	0	2	0	0
57	57	32	67.5	3	1	3	1	1
58	58	95	110.0	1	0	3	0	1
59	59	26	61.5	1	0	1	0	1
60	60	30	67.5	2	1	1	1	NA
61	61	22	58.5	1	2	1	0	0
62	62	NA	NA	1	0	2	0	1
63	63	32	67.0	2	0	3	0	1
64	64	32	67.0	2	0	2	1	1
65	65	31	66.0	1	2	1	1	0
66	66	35	72.0	2	0	2	0	1

12) Find the Missing Value for All Attributes.

```
colSums(is.na(Dataset))
```



```
64 colSums(is.na(Dataset))
65
66
64:1 (Top Level) ⚡ R Script

Console Terminal Background Jobs
R 4.2.2 ~ /

> colSums(is.na(Dataset))
      id      Age weight.kg Delivery_number Delivery_time      Blood      Heart Caesarian
      0         3         3             2             3         3         0         2
```

13) Find the Specific Row Number of Null Values of Age, Weight, Delivery Number, Delivery Time and Caesarian.

```
which(is.na(Dataset$Age))
```

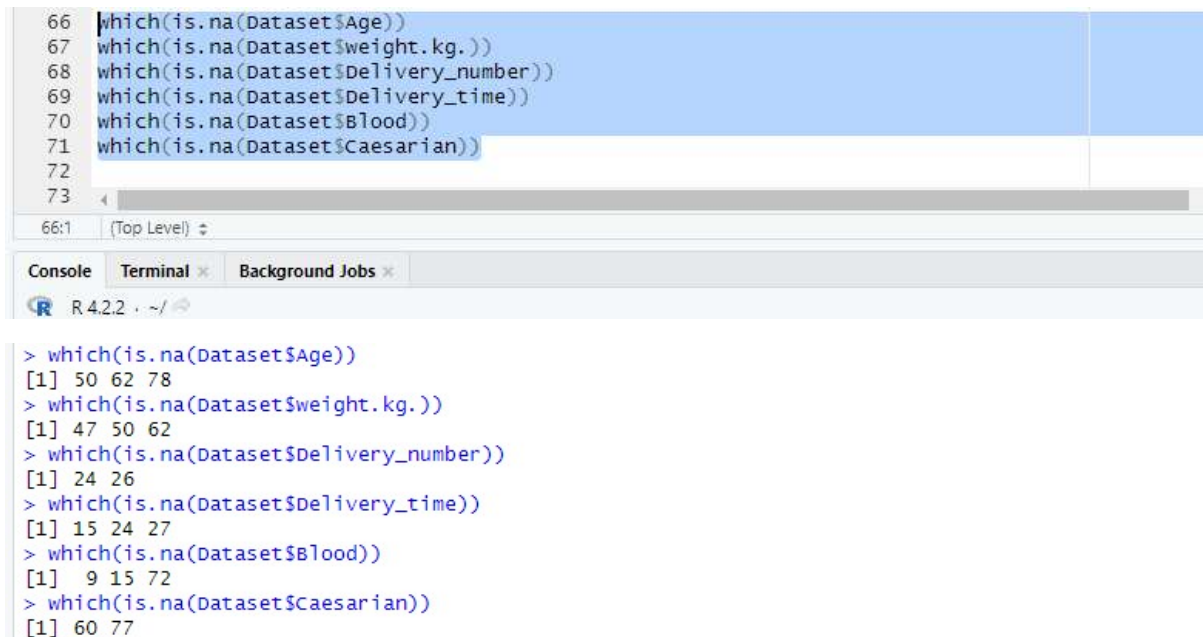
```
which(is.na(Dataset$weight.kg.))
```

```
which(is.na(Dataset$Delivery_number))
```

```
which(is.na(Dataset$Delivery_time))
```

```
which(is.na(Dataset$Blood))
```

```
which(is.na(Dataset$Caesarian))
```



```
66 which(is.na(Dataset$Age))
67 which(is.na(Dataset$weight.kg.))
68 which(is.na(Dataset$Delivery_number))
69 which(is.na(Dataset$Delivery_time))
70 which(is.na(Dataset$Blood))
71 which(is.na(Dataset$Caesarian))
72
73
66:1 (Top Level) ⚡ R Script

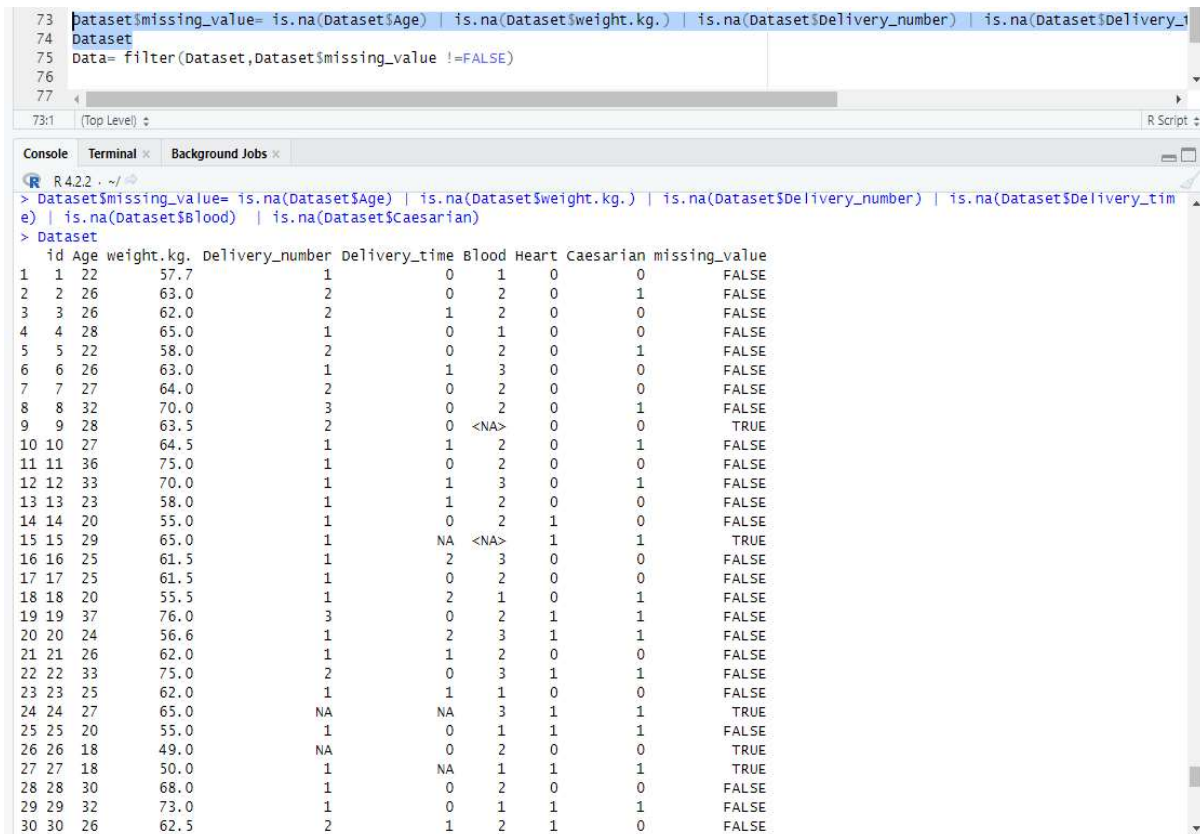
Console Terminal Background Jobs
R 4.2.2 ~ /

> which(is.na(Dataset$Age))
[1] 50 62 78
> which(is.na(Dataset$weight.kg.))
[1] 47 50 62
> which(is.na(Dataset$Delivery_number))
[1] 24 26
> which(is.na(Dataset$Delivery_time))
[1] 15 24 27
> which(is.na(Dataset$Blood))
[1] 9 15 72
> which(is.na(Dataset$Caesarian))
[1] 60 77
```

14) Detect the Outlier as a Missing Value.

```
Dataset$missing_value= is.na(Dataset$Age) | is.na(Dataset$weight.kg.) |  
is.na(Dataset$Delivery_number) | is.na(Dataset$Delivery_time) | is.na(Dataset$Blood) |  
is.na(Dataset$Caesarian)
```

Dataset



```
73 Dataset$missing_value= is.na(Dataset$Age) | is.na(Dataset$weight.kg.) | is.na(Dataset$Delivery_number) | is.na(Dataset$Delivery_t  
74 Dataset  
75 Data= filter(Dataset, Dataset$missing_value !=FALSE)  
76  
77
```

Console

```
> Dataset$missing_value= is.na(Dataset$Age) | is.na(Dataset$weight.kg.) | is.na(Dataset$Delivery_number) | is.na(Dataset$Delivery_tim  
> | is.na(Dataset$Blood) | is.na(Dataset$Caesarian)  
> Dataset
```

	id	Age	weight.kg.	Delivery_number	Delivery_time	Blood	Heart	Caesarian	missing_value
1	1	22	57.7	1	0	1	0	0	FALSE
2	2	26	63.0	2	0	2	0	1	FALSE
3	3	26	62.0	2	1	2	0	0	FALSE
4	4	28	65.0	1	0	1	0	0	FALSE
5	5	22	58.0	2	0	2	0	1	FALSE
6	6	26	63.0	1	1	3	0	0	FALSE
7	7	27	64.0	2	0	2	0	0	FALSE
8	8	32	70.0	3	0	2	0	1	FALSE
9	9	28	63.5	2	0	<NA>	0	0	TRUE
10	10	27	64.5	1	1	2	0	1	FALSE
11	11	36	75.0	1	0	2	0	0	FALSE
12	12	33	70.0	1	1	3	0	1	FALSE
13	13	23	58.0	1	1	2	0	0	FALSE
14	14	20	55.0	1	0	2	1	0	FALSE
15	15	29	65.0	1	NA	<NA>	1	1	TRUE
16	16	25	61.5	1	2	3	0	0	FALSE
17	17	25	61.5	1	0	2	0	0	FALSE
18	18	20	55.5	1	2	1	0	1	FALSE
19	19	37	76.0	3	0	2	1	1	FALSE
20	20	24	56.6	1	2	3	1	1	FALSE
21	21	26	62.0	1	1	2	0	0	FALSE
22	22	33	75.0	2	0	3	1	1	FALSE
23	23	25	62.0	1	1	1	0	0	FALSE
24	24	27	65.0	NA	NA	3	1	1	TRUE
25	25	20	55.0	1	0	1	1	1	FALSE
26	26	18	49.0	NA	0	2	0	0	TRUE
27	27	18	50.0	1	NA	1	1	1	TRUE
28	28	30	68.0	1	0	2	0	0	FALSE
29	29	32	73.0	1	0	1	1	1	FALSE
30	30	26	62.5	2	1	2	1	0	FALSE

```
Data= filter(Dataset, Dataset$missing_value !=FALSE)
```

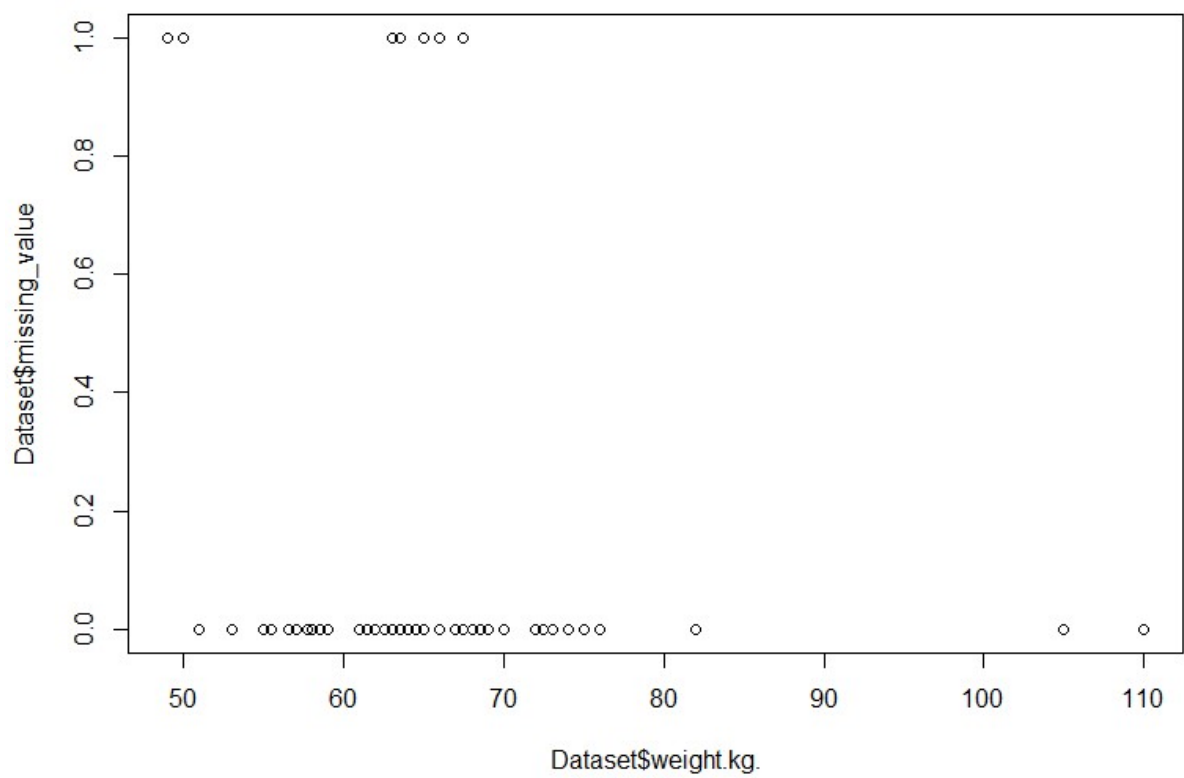
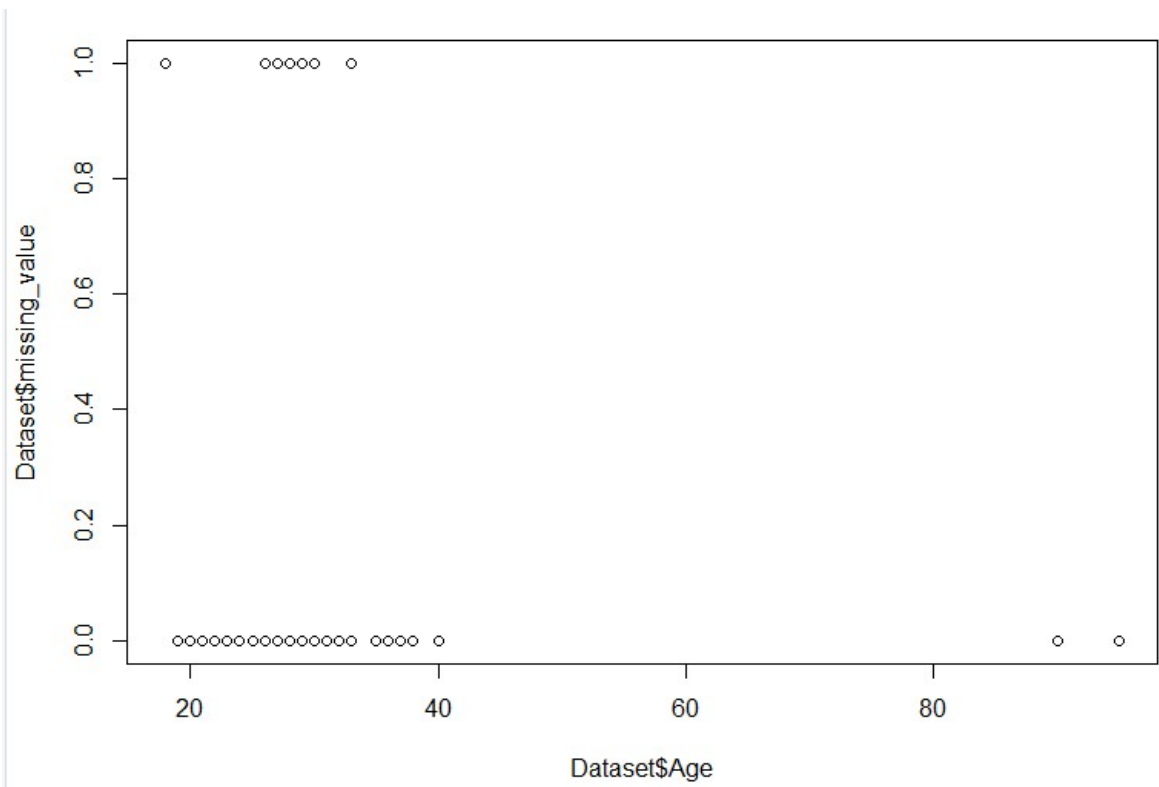
```
plot(Dataset$Age, Dataset$missing_value)
```

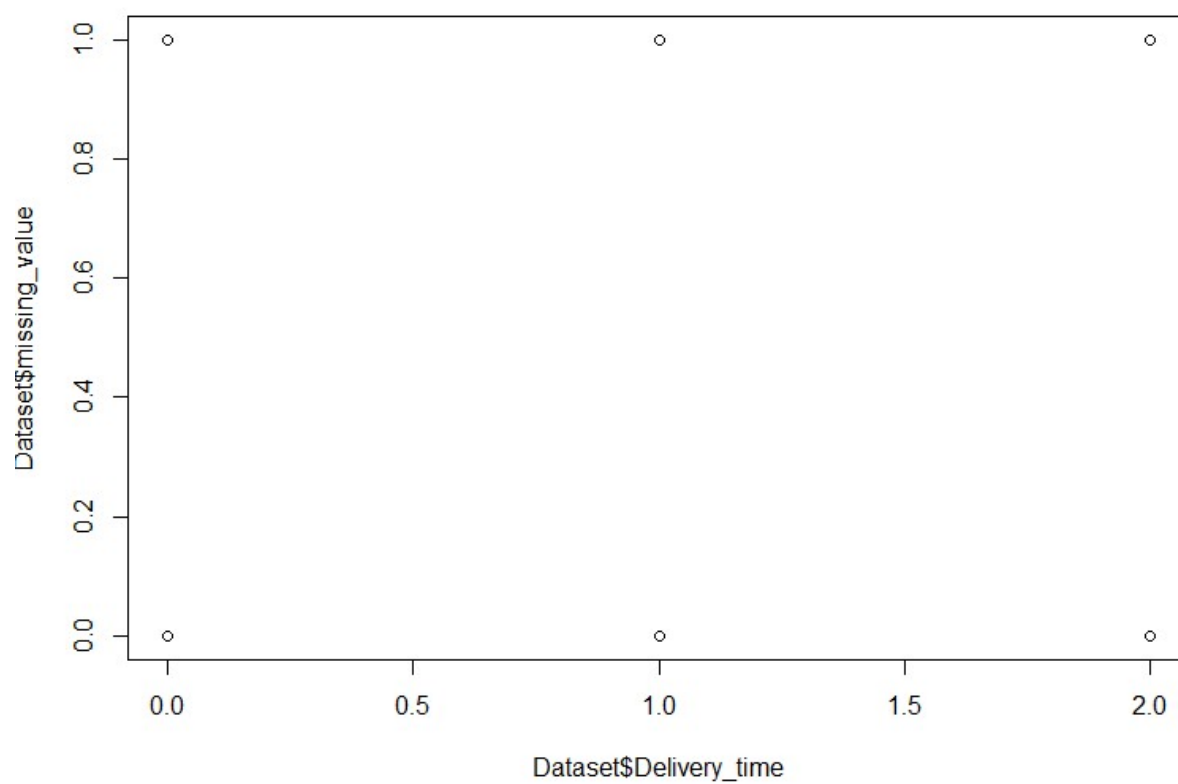
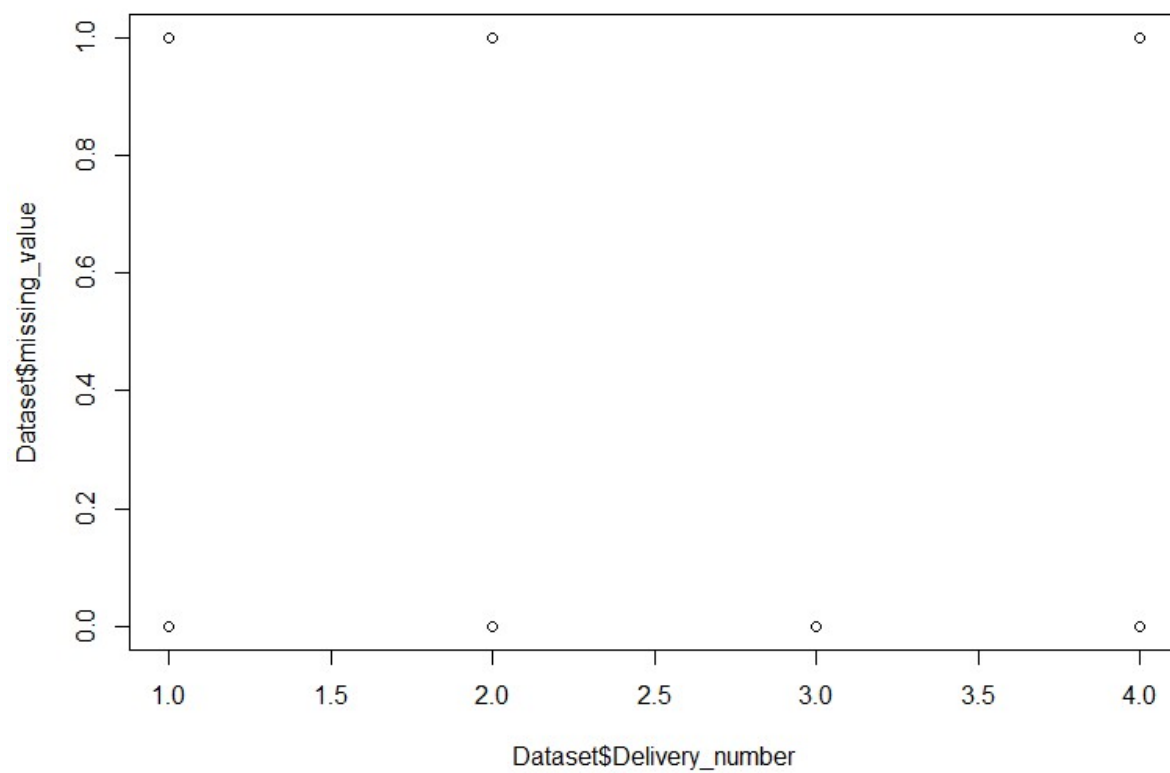
```
plot(Dataset$weight.kg., Dataset$missing_value)
```

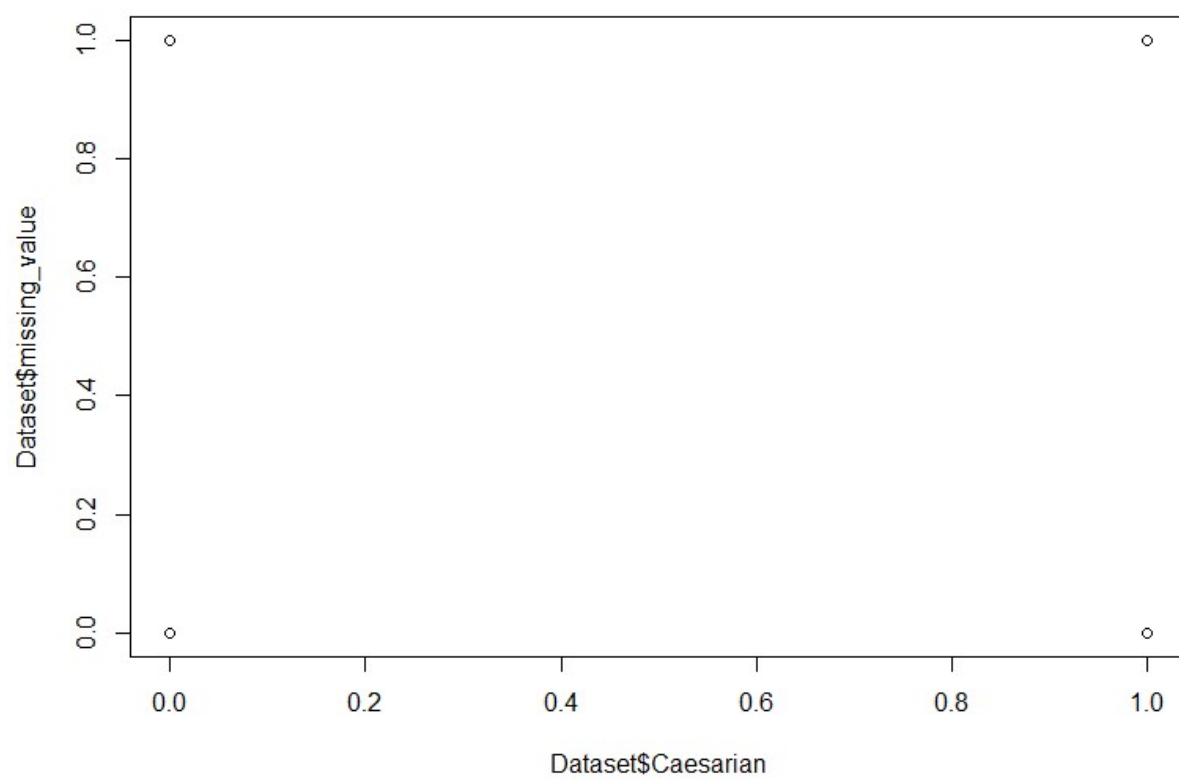
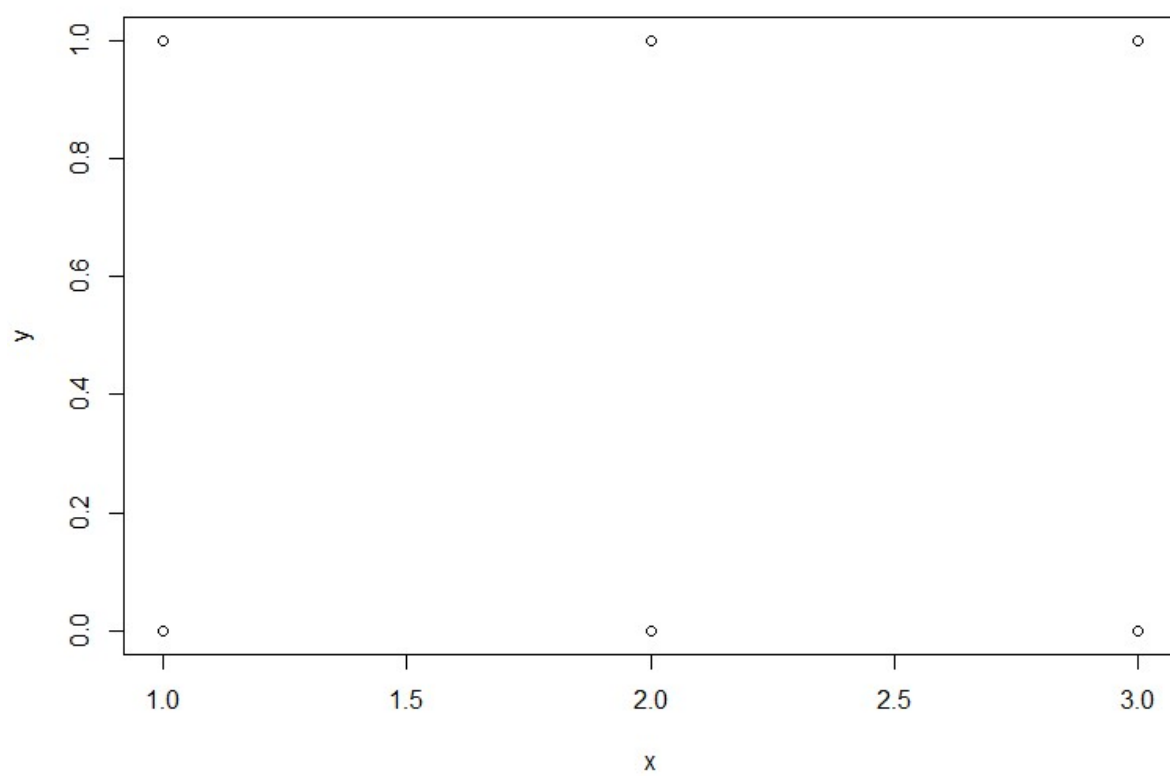
```
plot(Dataset$Delivery_number, Dataset$missing_value)
```

```
plot(Dataset$Delivery_number, Dataset$Blood)
```

```
plot(Dataset$Caesarian, Dataset$missing_value)
```







15) Recover missing values by the following strategies for Age, Weight, Delivery Number, Delivery Time, Blood, and Caesarian attributes.

I. Recover missing values with the mean value.

```
meanvalueAge <-mean(Dataset$Age,na.rm=TRUE)
```

```
meanvalueAge
```

```
Dataset[is.na(Dataset$Age), "Age"] <-meanvalueAge
```

```
Dataset
```

93	meanvalueAge <-mean(Dataset\$Age,na.rm=TRUE)	
94	meanvalueAge	
95	Dataset[is.na(Dataset\$Age), "Age"] <-meanvalueAge	
96	Dataset	
97		
98		
99		

93:1	(Top Level) ⌵
------	---------------

Console	Terminal x	Background Jobs x
---------	------------	-------------------

R 4.2.2 · ~/

```
> meanvalueAge <-mean(Dataset$Age,na.rm=TRUE)
> meanvalueAge
[1] 29.67532
> Dataset[is.na(Dataset$Age), "Age"] <-meanvalueAge
> Dataset
```

	id	Age	weight.kg.	Delivery_number	Delivery_time	Blood	Heart	Caesarian	missing_value
1	1	22.00000	57.7	1	0	1	0	0	FALSE
2	2	26.00000	63.0	2	0	2	0	1	FALSE
3	3	26.00000	62.0	2	1	2	0	0	FALSE
4	4	28.00000	65.0	1	0	1	0	0	FALSE
5	5	22.00000	58.0	2	0	2	0	1	FALSE
6	6	26.00000	63.0	1	1	3	0	0	FALSE
7	7	27.00000	64.0	2	0	2	0	0	FALSE
8	8	32.00000	70.0	3	0	2	0	1	FALSE
9	9	28.00000	63.5	2	0	<NA>	0	0	TRUE
10	10	27.00000	64.5	1	1	2	0	1	FALSE
11	11	36.00000	75.0	1	0	2	0	0	FALSE
12	12	33.00000	70.0	1	1	3	0	1	FALSE
13	13	23.00000	58.0	1	1	2	0	0	FALSE
14	14	20.00000	55.0	1	0	2	1	0	FALSE
15	15	29.00000	65.0	1	NA	<NA>	1	1	TRUE
16	16	25.00000	61.5	1	2	3	0	0	FALSE
17	17	25.00000	61.5	1	0	2	0	0	FALSE
18	18	20.00000	55.5	1	2	1	0	1	FALSE
19	19	37.00000	76.0	3	0	2	1	1	FALSE
20	20	24.00000	56.6	1	2	3	1	1	FALSE
21	21	26.00000	62.0	1	1	2	0	0	FALSE
22	22	33.00000	75.0	2	0	3	1	1	FALSE
23	23	25.00000	62.0	1	1	1	0	0	FALSE
24	24	27.00000	65.0	NA	NA	3	1	1	TRUE
25	25	20.00000	55.0	1	0	1	1	1	FALSE
26	26	18.00000	49.0	NA	0	2	0	0	TRUE
27	27	18.00000	50.0	1	NA	1	1	1	TRUE
28	28	30.00000	68.0	1	0	2	0	0	FALSE

```
meanvalueWeight <-mean(Dataset$weight.kg.,na.rm=TRUE)
```

```
meanvalueWeight
```

```
Dataset[is.na(Dataset$weight.kg.), "weight.kg."] <-meanvalueWeight
```

```
Dataset
```

The screenshot shows an R Studio interface with a script editor and a console. The script editor contains R code for data cleaning and analysis. The console shows the execution of the code, including the calculation of the mean weight and the resulting data table.

```
85 install.packages("matrixStats")
86 library(matrixStats)
87 install.packages("dplyr")
88 library(dplyr)
89
90 remove_missingvalue<- na.omit(Dataset)
91 remove_missingvalue
92
93 meanvalueAge <-mean(Dataset$Age,na.rm=TRUE)
94 meanvalueAge
95 Dataset[is.na(Dataset$Age), "Age"] <-meanvalueAge
96 Dataset
97
98 meanvalueweight <-mean(Dataset$weight.kg.,na.rm=TRUE)
99 meanvalueweight
100 Dataset[is.na(Dataset$weight.kg.), "weight.kg."] <-meanvalueweight
101 Dataset
102
```

Console output:

```
> meanvalueweight
[1] 65.12727
> Dataset[is.na(Dataset$weight.kg.), "weight.kg."] <-meanvalueweight
> Dataset
```

	id	Age	weight.kg.	Delivery_number	Delivery_time	Blood	Heart	Caesarian	missing_value
1	1	22.00000	57.70000	1	0	1	0	0	FALSE
2	2	26.00000	63.00000	2	0	2	0	1	FALSE
3	3	26.00000	62.00000	2	1	2	0	0	FALSE
4	4	28.00000	65.00000	1	0	1	0	0	FALSE
5	5	22.00000	58.00000	2	0	2	0	1	FALSE
6	6	26.00000	63.00000	1	1	3	0	0	FALSE
7	7	27.00000	64.00000	2	0	2	0	0	FALSE
8	8	32.00000	70.00000	3	0	2	0	1	FALSE
9	9	28.00000	63.50000	2	0	<NA>	0	0	TRUE
10	10	27.00000	64.50000	1	1	2	0	1	FALSE
11	11	36.00000	75.00000	1	0	2	0	0	FALSE
12	12	33.00000	70.00000	1	1	3	0	1	FALSE
13	13	23.00000	58.00000	1	1	2	0	0	FALSE
14	14	20.00000	55.00000	1	0	2	1	0	FALSE
15	15	29.00000	65.00000	1	NA	<NA>	1	1	TRUE
16	16	25.00000	61.50000	1	2	3	0	0	FALSE
17	17	25.00000	61.50000	1	0	2	0	0	FALSE
18	18	20.00000	55.50000	1	2	1	0	1	FALSE
19	19	37.00000	76.00000	3	0	2	1	1	FALSE
20	20	24.00000	56.60000	1	2	3	1	1	FALSE
21	21	26.00000	62.00000	1	1	2	0	0	FALSE
22	22	33.00000	75.00000	2	0	3	1	1	FALSE
23	23	25.00000	62.00000	1	1	1	0	0	FALSE
24	24	27.00000	65.00000	NA	NA	3	1	1	TRUE
25	25	20.00000	55.00000	1	0	1	1	1	FALSE
26	26	18.00000	49.00000	NA	0	2	0	0	TRUE
27	27	18.00000	50.00000	1	NA	1	1	1	TRUE
28	28	30.00000	68.00000	1	0	2	0	0	FALSE
29	29	32.00000	73.00000	1	0	1	1	1	FALSE

```
meanvalueDelivery_number <-mean(Dataset$Delivery_number,na.rm=TRUE)
```

```
meanvalueDelivery_number
```

```
Dataset[is.na(Dataset$Delivery_number), "Delivery_number"] <-meanvalueDelivery_number
```

```
Dataset
```

103	meanvalueDelivery_number <-mean(Dataset\$Delivery_number,na.rm=TRUE)	
104	meanvalueDelivery_number	
105	Dataset[is.na(Dataset\$Delivery_number), "Delivery_number"] <-meanvalueDelivery_number	
106	Dataset	
107		

103:1 (Top Level) ⚙

Console	Terminal	Background Jobs
R 4.2.2 . ~/		
80 80 24.00000 57.00000 2 2 2 0 0 FALSE		
> meanvalueDelivery_number <-mean(Dataset\$Delivery_number,na.rm=TRUE)		
> meanvalueDelivery_number		
[1] 1.679487		
> Dataset[is.na(Dataset\$Delivery_number), "Delivery_number"] <-meanvalueDelivery_number		
> Dataset		
	id	Age weight.kg. Delivery_number Delivery_time Blood Heart Caesarian missing_value
1	1	22.00000 57.70000 1.000000 0 1 0 0 FALSE
2	2	26.00000 63.00000 2.000000 0 2 0 1 FALSE
3	3	26.00000 62.00000 2.000000 1 2 0 0 FALSE
4	4	28.00000 65.00000 1.000000 0 1 0 0 FALSE
5	5	22.00000 58.00000 2.000000 0 2 0 1 FALSE
6	6	26.00000 63.00000 1.000000 1 3 0 0 FALSE
7	7	27.00000 64.00000 2.000000 0 2 0 0 FALSE
8	8	32.00000 70.00000 3.000000 0 2 0 1 FALSE
9	9	28.00000 63.50000 2.000000 0 <NA> 0 0 TRUE
10	10	27.00000 64.50000 1.000000 1 2 0 1 FALSE
11	11	36.00000 75.00000 1.000000 0 2 0 0 FALSE
12	12	33.00000 70.00000 1.000000 1 3 0 1 FALSE
13	13	23.00000 58.00000 1.000000 1 2 0 0 FALSE
14	14	20.00000 55.00000 1.000000 0 2 1 0 FALSE
15	15	29.00000 65.00000 1.000000 NA <NA> 1 1 TRUE
16	16	25.00000 61.50000 1.000000 2 3 0 0 FALSE
17	17	25.00000 61.50000 1.000000 0 2 0 0 FALSE
18	18	20.00000 55.50000 1.000000 2 1 0 1 FALSE
19	19	37.00000 76.00000 3.000000 0 2 1 1 FALSE
20	20	24.00000 56.60000 1.000000 2 3 1 1 FALSE
21	21	26.00000 62.00000 1.000000 1 2 0 0 FALSE
22	22	33.00000 75.00000 2.000000 0 3 1 1 FALSE
23	23	25.00000 62.00000 1.000000 1 1 0 0 FALSE
24	24	27.00000 65.00000 1.679487 NA 3 1 1 TRUE
25	25	20.00000 55.00000 1.000000 0 1 1 1 FALSE
26	26	18.00000 49.00000 1.679487 0 2 0 0 TRUE
27	27	18.00000 50.00000 1.000000 NA 1 1 1 TRUE


```
meanvalueDelivery_time <-mean(Dataset$Delivery_time,na.rm=TRUE)
```

```
meanvalueDelivery_time
```

```
Dataset[is.na(Dataset$Delivery_time), "Delivery_time"] <-meanvalueDelivery_time
```

```
Dataset
```

```
108 meanvalueDelivery_time <-mean(Dataset$Delivery_time,na.rm=TRUE)
109 meanvalueDelivery_time
110 Dataset[is.na(Dataset$Delivery_time), "Delivery_time"] <-meanvalueDelivery_time
111 Dataset
112
113
```

108:1 (Top Level) ▾

Console Terminal Background Jobs

R 4.2.2 . ~/

```
> meanvalueDelivery_time
[1] 0.6233766
> Dataset[is.na(Dataset$Delivery_time), "Delivery_time"] <-meanvalueDelivery_time
> Dataset
```

	id	Age	weight.kg.	Delivery_number	Delivery_time	Blood	Heart	Caesarian	missing_value
1	1	22.00000	57.70000	1.000000	0.0000000	1	0	0	FALSE
2	2	26.00000	63.00000	2.000000	0.0000000	2	0	1	FALSE
3	3	26.00000	62.00000	2.000000	1.0000000	2	0	0	FALSE
4	4	28.00000	65.00000	1.000000	0.0000000	1	0	0	FALSE
5	5	22.00000	58.00000	2.000000	0.0000000	2	0	1	FALSE
6	6	26.00000	63.00000	1.000000	1.0000000	3	0	0	FALSE
7	7	27.00000	64.00000	2.000000	0.0000000	2	0	0	FALSE
8	8	32.00000	70.00000	3.000000	0.0000000	2	0	1	FALSE
9	9	28.00000	63.50000	2.000000	0.0000000	<NA>	0	0	TRUE
10	10	27.00000	64.50000	1.000000	1.0000000	2	0	1	FALSE
11	11	36.00000	75.00000	1.000000	0.0000000	2	0	0	FALSE
12	12	33.00000	70.00000	1.000000	1.0000000	3	0	1	FALSE
13	13	23.00000	58.00000	1.000000	1.0000000	2	0	0	FALSE
14	14	20.00000	55.00000	1.000000	0.0000000	2	1	0	FALSE
15	15	29.00000	65.00000	1.000000	0.6233766	<NA>	1	1	TRUE
16	16	25.00000	61.50000	1.000000	2.0000000	3	0	0	FALSE
17	17	25.00000	61.50000	1.000000	0.0000000	2	0	0	FALSE
18	18	20.00000	55.50000	1.000000	2.0000000	1	0	1	FALSE
19	19	37.00000	76.00000	3.000000	0.0000000	2	1	1	FALSE
20	20	24.00000	56.60000	1.000000	2.0000000	3	1	1	FALSE
21	21	26.00000	62.00000	1.000000	1.0000000	2	0	0	FALSE
22	22	33.00000	75.00000	2.000000	0.0000000	3	1	1	FALSE
23	23	25.00000	62.00000	1.000000	1.0000000	1	0	0	FALSE
24	24	27.00000	65.00000	1.679487	0.6233766	3	1	1	TRUE
25	25	20.00000	55.00000	1.000000	0.0000000	1	1	1	FALSE
26	26	18.00000	49.00000	1.679487	0.0000000	2	0	0	TRUE
27	27	18.00000	50.00000	1.000000	0.6233766	1	1	1	TRUE
28	28	30.00000	68.00000	1.000000	0.0000000	2	0	0	FALSE
29	29	32.00000	73.00000	1.000000	0.0000000	1	1	1	FALSE

```
meanvalueCaesarian <-mean(Dataset$Caesarian,na.rm=TRUE)
```

```
meanvalueCaesarian
```

```
Dataset[is.na(Dataset$Caesarian), "Caesarian"] <-meanvalueCaesarian
```

```
Dataset
```

20-42970-1_IDS_MidProject.R
Dataset
Source on Save
112
113 meanvalueCaesarian <-mean(Dataset\$Caesarian,na.rm=TRUE)
114 meanvalueCaesarian
115 Dataset[is.na(Dataset\$Caesarian), "Caesarian"] <-meanvalueCaesarian
116 Dataset
117
93:1 (Top Level)

Console
Terminal
Background Jobs
R 4.2.2 . ~/
> meanvalueCaesarian
[1] 0.5641026
> Dataset[is.na(Dataset\$Caesarian), "Caesarian"] <-meanvalueCaesarian
> Dataset

	id	Age	weight.kg.	Delivery_number	Delivery_time	Blood	Heart	Caesarian	missing_value
1	1	22.00000	57.70000	1.000000	0.0000000	1	0	0.0000000	FALSE
2	2	26.00000	63.00000	2.000000	0.0000000	2	0	1.0000000	FALSE
3	3	26.00000	62.00000	2.000000	1.0000000	2	0	0.0000000	FALSE
4	4	28.00000	65.00000	1.000000	0.0000000	1	0	0.0000000	FALSE
5	5	22.00000	58.00000	2.000000	0.0000000	2	0	1.0000000	FALSE
6	6	26.00000	63.00000	1.000000	1.0000000	3	0	0.0000000	FALSE
7	7	27.00000	64.00000	2.000000	0.0000000	2	0	0.0000000	FALSE
8	8	32.00000	70.00000	3.000000	0.0000000	2	0	1.0000000	FALSE
9	9	28.00000	63.50000	2.000000	0.0000000	<NA>	0	0.0000000	TRUE
10	10	27.00000	64.50000	1.000000	1.0000000	2	0	1.0000000	FALSE
11	11	36.00000	75.00000	1.000000	0.0000000	2	0	0.0000000	FALSE
12	12	33.00000	70.00000	1.000000	1.0000000	3	0	1.0000000	FALSE
13	13	23.00000	58.00000	1.000000	1.0000000	2	0	0.0000000	FALSE
14	14	20.00000	55.00000	1.000000	0.0000000	2	1	0.0000000	FALSE
15	15	29.00000	65.00000	1.000000	0.6233766	<NA>	1	1.0000000	TRUE
16	16	25.00000	61.50000	1.000000	2.0000000	3	0	0.0000000	FALSE
17	17	25.00000	61.50000	1.000000	0.0000000	2	0	0.0000000	FALSE
18	18	20.00000	55.50000	1.000000	2.0000000	1	0	1.0000000	FALSE
19	19	37.00000	76.00000	3.000000	0.0000000	2	1	1.0000000	FALSE
20	20	24.00000	56.60000	1.000000	2.0000000	3	1	1.0000000	FALSE
21	21	26.00000	62.00000	1.000000	1.0000000	2	0	0.0000000	FALSE
22	22	33.00000	75.00000	2.000000	0.0000000	3	1	1.0000000	FALSE
23	23	25.00000	62.00000	1.000000	1.0000000	1	0	0.0000000	FALSE
24	24	27.00000	65.00000	1.679487	0.6233766	3	1	1.0000000	TRUE
25	25	20.00000	55.00000	1.000000	0.0000000	1	1	1.0000000	FALSE
26	26	18.00000	49.00000	1.679487	0.0000000	2	0	0.0000000	TRUE
27	27	18.00000	50.00000	1.000000	0.6233766	1	1	1.0000000	TRUE
28	28	30.00000	68.00000	1.000000	0.0000000	2	0	0.0000000	FALSE
29	29	32.00000	73.00000	1.000000	0.0000000	1	1	1.0000000	FALSE
30	30	26.00000	62.50000	2.000000	1.0000000	2	1	0.0000000	FALSE
31	31	25.00000	58.00000	1.000000	0.0000000	3	0	0.0000000	FALSE
32	32	40.00000	82.00000	1.000000	0.0000000	2	1	1.0000000	FALSE
33	33	32.00000	68.00000	2.000000	0.0000000	1	1	1.0000000	FALSE
34	34	27.00000	63.00000	2.000000	0.0000000	2	1	1.0000000	FALSE
35	35	26.00000	59.00000	2.000000	2.0000000	2	0	1.0000000	FALSE
36	36	28.00000	66.00000	3.000000	0.0000000	1	0	1.0000000	FALSE
37	37	33.00000	75.00000	1.000000	1.0000000	2	0	0.0000000	FALSE
38	38	31.00000	69.00000	2.000000	2.0000000	2	0	0.0000000	FALSE
39	39	31.00000	63.00000	1.000000	0.0000000	2	0	0.0000000	FALSE
40	40	26.00000	59.00000	1.000000	2.0000000	3	1	1.0000000	FALSE
41	41	27.00000	63.00000	1.000000	0.0000000	1	1	1.0000000	FALSE
42	42	19.00000	51.00000	1.000000	0.0000000	2	0	1.0000000	FALSE

43	43	36.00000	73.00000	1.000000	1.0000000	1	0	1.0000000	FALSE
44	44	22.00000	57.00000	1.000000	0.0000000	2	0	1.0000000	FALSE
45	45	36.00000	72.50000	4.000000	0.0000000	1	1	1.0000000	FALSE
46	46	28.00000	62.50000	3.000000	0.0000000	2	1	1.0000000	FALSE
47	47	26.00000	65.12727	1.000000	0.0000000	2	0	0.0000000	TRUE
48	48	32.00000	67.50000	2.000000	0.0000000	1	1	1.0000000	FALSE
49	49	26.00000	62.50000	2.000000	2.0000000	2	0	0.0000000	FALSE
50	50	29.67532	65.12727	2.000000	0.0000000	3	1	1.0000000	TRUE
51	51	33.00000	68.50000	3.000000	2.0000000	2	1	0.0000000	FALSE
52	52	21.00000	53.00000	2.000000	1.0000000	3	1	1.0000000	FALSE
53	53	30.00000	68.00000	3.000000	2.0000000	1	0	0.0000000	FALSE
54	54	35.00000	74.00000	1.000000	1.0000000	3	0	0.0000000	FALSE
55	55	29.00000	63.50000	2.000000	0.0000000	2	1	1.0000000	FALSE
56	56	25.00000	59.00000	2.000000	0.0000000	2	0	0.0000000	FALSE
57	57	32.00000	67.50000	3.000000	1.0000000	3	1	1.0000000	FALSE
58	58	95.00000	110.00000	1.000000	0.0000000	3	0	1.0000000	FALSE
59	59	26.00000	61.50000	1.000000	0.0000000	1	0	1.0000000	FALSE
60	60	30.00000	67.50000	2.000000	1.0000000	1	1	0.5641026	TRUE
61	61	22.00000	58.50000	1.000000	2.0000000	1	0	0.0000000	FALSE
62	62	29.67532	65.12727	1.000000	0.0000000	2	0	1.0000000	TRUE
63	63	32.00000	67.00000	2.000000	0.0000000	3	0	1.0000000	FALSE
64	64	32.00000	67.00000	2.000000	0.0000000	2	1	1.0000000	FALSE
65	65	31.00000	66.00000	1.000000	2.0000000	1	1	0.0000000	FALSE
66	66	35.00000	72.00000	2.000000	0.0000000	2	0	1.0000000	FALSE
67	67	28.00000	62.50000	3.000000	0.0000000	2	0	1.0000000	FALSE
68	68	29.00000	64.50000	2.000000	0.0000000	2	1	0.0000000	FALSE
69	69	25.00000	62.00000	1.000000	0.0000000	3	0	1.0000000	FALSE
70	70	27.00000	61.00000	2.000000	2.0000000	3	0	0.0000000	FALSE
71	71	90.00000	105.00000	1.000000	0.0000000	3	0	1.0000000	FALSE
72	72	29.00000	65.00000	1.000000	2.0000000	<NA>	1	1.0000000	TRUE
73	73	28.00000	64.00000	2.000000	0.0000000	2	0	0.0000000	FALSE
74	74	32.00000	69.00000	3.000000	0.0000000	2	1	0.0000000	FALSE
75	75	38.00000	75.00000	3.000000	2.0000000	1	1	1.0000000	FALSE
76	76	27.00000	62.50000	2.000000	1.0000000	2	0	0.0000000	FALSE
77	77	33.00000	66.00000	4.000000	0.0000000	2	0	0.5641026	TRUE
78	78	29.67532	63.00000	2.000000	1.0000000	1	0	1.0000000	TRUE
79	79	25.00000	58.00000	1.000000	2.0000000	3	0	1.0000000	FALSE
80	80	24.00000	57.00000	2.000000	2.0000000	2	0	0.0000000	FALSE

```
# Replace "NA" with NA in the 'Sex' column
```

```
Dataset$Blood[Dataset$Blood == "NA"] <- NA
```

```
# Convert the 'Sex' column to character
```

```
Dataset$Blood <- as.character(Dataset$Blood)
```

```
# Calculate the mean excluding NAs
```

```
meanvalueBlood <- mean(as.numeric(Dataset$Blood), na.rm = TRUE)
```

```
# Replace missing values in 'Sex' column with the mean
```

```
Dataset$Blood[is.na(Dataset$Blood)] <- as.character(meanvalueBlood)
```

```
Dataset
```

```

> Dataset$Blood[Dataset$Blood == "NA"] <- NA
> Dataset$Blood <- as.character(Dataset$Blood)
>
> # Calculate the mean excluding NAs
> meanvalueBlood <- mean(as.numeric(Dataset$Blood), na.rm = TRUE)
> Dataset$Blood[is.na(Dataset$Blood)] <- as.character(meanvalueBlood)
> Dataset

```

	id	Age	weight.kg.	Delivery_number	Delivery_time	Blood	Heart	Caesarian	missing_value
1	1	22.00000	57.70000	1.000000	0.000000	1	0	0.000000	FALSE
2	2	26.00000	63.00000	2.000000	0.000000	2	0	1.000000	FALSE
3	3	26.00000	62.00000	2.000000	1.000000	2	0	0.000000	FALSE
4	4	28.00000	65.00000	1.000000	0.000000	1	0	0.000000	FALSE
5	5	22.00000	58.00000	2.000000	0.000000	2	0	1.000000	FALSE
6	6	26.00000	63.00000	1.000000	1.000000	3	0	0.000000	FALSE
7	7	27.00000	64.00000	2.000000	0.000000	2	0	0.000000	FALSE
8	8	32.00000	70.00000	3.000000	0.000000	2	0	1.000000	FALSE
9	9	28.00000	63.50000	2.000000	0.000000	1.97402597402597	0	0.000000	TRUE
10	10	27.00000	64.50000	1.000000	1.000000	2	0	1.000000	FALSE
11	11	36.00000	75.00000	1.000000	0.000000	2	0	0.000000	FALSE
12	12	33.00000	70.00000	1.000000	1.000000	3	0	1.000000	FALSE
13	13	23.00000	58.00000	1.000000	1.000000	2	0	0.000000	FALSE
14	14	20.00000	55.00000	1.000000	0.000000	2	1	0.000000	FALSE
15	15	29.00000	65.00000	1.000000	0.6233766	1.97402597402597	1	1.000000	TRUE
16	16	25.00000	61.50000	1.000000	2.000000	3	0	0.000000	FALSE
17	17	25.00000	61.50000	1.000000	0.000000	2	0	0.000000	FALSE
18	18	20.00000	55.50000	1.000000	2.000000	1	0	1.000000	FALSE
19	19	37.00000	76.00000	3.000000	0.000000	2	1	1.000000	FALSE
20	20	24.00000	56.60000	1.000000	2.000000	3	1	1.000000	FALSE
21	21	26.00000	62.00000	1.000000	1.000000	2	0	0.000000	FALSE
22	22	33.00000	75.00000	2.000000	0.000000	3	1	1.000000	FALSE
23	23	25.00000	62.00000	1.000000	1.000000	1	0	0.000000	FALSE
24	24	27.00000	65.00000	1.679487	0.6233766	3	1	1.000000	TRUE
25	25	20.00000	55.00000	1.000000	0.000000	1	1	1.000000	FALSE
26	26	18.00000	49.00000	1.679487	0.000000	2	0	0.000000	TRUE
27	27	18.00000	50.00000	1.000000	0.6233766	1	1	1.000000	TRUE
28	28	30.00000	68.00000	1.000000	0.000000	2	0	0.000000	FALSE
29	29	32.00000	73.00000	1.000000	0.000000	1	1	1.000000	FALSE
30	30	26.00000	62.50000	2.000000	1.000000	2	1	0.000000	FALSE
31	31	25.00000	58.00000	1.000000	0.000000	3	0	0.000000	FALSE
32	32	40.00000	82.00000	1.000000	0.000000	2	1	1.000000	FALSE
33	33	32.00000	68.00000	2.000000	0.000000	1	1	1.000000	FALSE
34	34	27.00000	63.00000	2.000000	0.000000	2	1	1.000000	FALSE
35	35	26.00000	59.00000	2.000000	2.000000	2	0	1.000000	FALSE
36	36	28.00000	66.00000	3.000000	0.000000	1	0	1.000000	FALSE
37	37	33.00000	75.00000	1.000000	1.000000	2	0	0.000000	FALSE
38	38	31.00000	69.00000	2.000000	2.000000	2	0	0.000000	FALSE
39	39	31.00000	63.00000	1.000000	0.000000	2	0	0.000000	FALSE
40	40	26.00000	59.00000	1.000000	2.000000	3	1	1.000000	FALSE
41	41	27.00000	63.00000	1.000000	0.000000	1	1	1.000000	FALSE
42	42	19.00000	51.00000	1.000000	0.000000	2	0	1.000000	FALSE
43	43	36.00000	73.00000	1.000000	1.000000	1	0	1.000000	FALSE
44	44	22.00000	57.00000	1.000000	0.000000	2	0	1.000000	FALSE
45	45	36.00000	72.50000	4.000000	0.000000	1	1	1.000000	FALSE
46	46	28.00000	62.50000	3.000000	0.000000	2	1	1.000000	FALSE
47	47	26.00000	65.12727	1.000000	0.000000	2	0	0.000000	TRUE
48	48	32.00000	67.50000	2.000000	0.000000	1	1	1.000000	FALSE
49	49	26.00000	62.50000	2.000000	2.000000	2	0	0.000000	FALSE
50	50	29.67532	65.12727	2.000000	0.000000	3	1	1.000000	TRUE
51	51	33.00000	68.50000	3.000000	2.000000	2	1	0.000000	FALSE
52	52	21.00000	53.00000	2.000000	1.000000	3	1	1.000000	FALSE
53	53	30.00000	68.00000	3.000000	2.000000	1	0	0.000000	FALSE
54	54	35.00000	74.00000	1.000000	1.000000	3	0	0.000000	FALSE
55	55	29.00000	63.50000	2.000000	0.000000	2	1	1.000000	FALSE
56	56	25.00000	59.00000	2.000000	0.000000	2	0	0.000000	FALSE
57	57	32.00000	67.50000	3.000000	1.000000	3	1	1.000000	FALSE
58	58	95.00000	110.00000	1.000000	0.000000	3	0	1.000000	FALSE
59	59	26.00000	61.50000	1.000000	0.000000	1	0	1.000000	FALSE
60	60	30.00000	67.50000	2.000000	1.000000	1	1	0.5641026	TRUE
61	61	22.00000	58.50000	1.000000	2.000000	1	0	0.000000	FALSE
62	62	29.67532	65.12727	1.000000	0.000000	2	0	1.000000	TRUE
63	63	32.00000	67.00000	2.000000	0.000000	3	0	1.000000	FALSE
64	64	32.00000	67.00000	2.000000	0.000000	2	1	1.000000	FALSE
65	65	31.00000	66.00000	1.000000	2.000000	1	1	0.000000	FALSE
66	66	35.00000	72.00000	2.000000	0.000000	2	0	1.000000	FALSE
67	67	28.00000	62.50000	3.000000	0.000000	2	0	1.000000	FALSE
68	68	29.00000	64.50000	2.000000	0.000000	2	1	0.000000	FALSE
69	69	25.00000	62.00000	1.000000	0.000000	3	0	1.000000	FALSE
70	70	27.00000	61.00000	2.000000	2.000000	3	0	0.000000	FALSE
71	71	90.00000	105.00000	1.000000	0.000000	3	0	1.000000	FALSE
72	72	29.00000	65.00000	1.000000	2.000000	1.97402597402597	1	1.000000	TRUE
73	73	28.00000	64.00000	2.000000	0.000000	2	0	0.000000	FALSE
74	74	32.00000	69.00000	3.000000	0.000000	2	1	0.000000	FALSE
75	75	38.00000	75.00000	3.000000	2.000000	1	1	1.000000	FALSE
76	76	27.00000	62.50000	2.000000	1.000000	2	0	0.000000	FALSE
77	77	33.00000	66.00000	4.000000	0.000000	2	0	0.5641026	TRUE
78	78	29.67532	63.00000	2.000000	1.000000	1	0	1.000000	TRUE
79	79	25.00000	58.00000	1.000000	2.000000	3	0	1.000000	FALSE
80	80	24.00000	57.00000	2.000000	2.000000	2	0	0.000000	FALSE

II. Recover missing values with the median value.

```
medianvalueAge <-median(Dataset$Age,na.rm=TRUE)
```

```
medianvalueAge
```

```
Dataset[is.na(Dataset$Age), "Age"] <-medianvalueAge
```

```
Dataset
```

```
120 medianvalueAge <-median(Dataset$Age,na.rm=TRUE)
121 medianvalueAge
122 Dataset[is.na(Dataset$Age), "Age"] <-medianvalueAge
123 Dataset
```

123:8 (Top Level) ↕

Console	Terminal ×	Background Jobs ×
R 4.2.2 . ~/		
80 80 24 57.0 2 2 2 0 0 FALSE		
>		
> Data= filter(Dataset,Dataset\$missing_value !=FALSE)		
> medianvalueAge <-median(Dataset\$Age,na.rm=TRUE)		
> medianvalueAge		
[1] 28		
> Dataset[is.na(Dataset\$Age), "Age"] <-medianvalueAge		
> Dataset		
	id	Age weight.kg. Delivery_number Delivery_time Blood Heart Caesarian missing_value
1	1	22 57.7 1 0 1 0 0 FALSE
2	2	26 63.0 2 0 2 0 1 FALSE
3	3	26 62.0 2 1 2 0 0 FALSE
4	4	28 65.0 1 0 1 0 0 FALSE
5	5	22 58.0 2 0 2 0 1 FALSE
6	6	26 63.0 1 1 3 0 0 FALSE
7	7	27 64.0 2 0 2 0 0 FALSE
8	8	32 70.0 3 0 2 0 1 FALSE
9	9	28 63.5 2 0 <NA> 0 0 TRUE
10	10	27 64.5 1 1 2 0 1 FALSE
11	11	36 75.0 1 0 2 0 0 FALSE
12	12	33 70.0 1 1 3 0 1 FALSE
13	13	23 58.0 1 1 2 0 0 FALSE
14	14	20 55.0 1 0 2 1 0 FALSE
15	15	29 65.0 1 NA <NA> 1 1 TRUE
16	16	25 61.5 1 2 3 0 0 FALSE
17	17	25 61.5 1 0 2 0 0 FALSE
18	18	20 55.5 1 2 1 0 1 FALSE
19	19	37 76.0 3 0 2 1 1 FALSE
20	20	24 56.6 1 2 3 1 1 FALSE
21	21	26 62.0 1 1 2 0 0 FALSE
22	22	33 75.0 2 0 3 1 1 FALSE
23	23	25 62.0 1 1 1 0 0 FALSE
24	24	27 65.0 NA NA 3 1 1 TRUE
25	25	20 55.0 1 0 1 1 1 FALSE

```
medianvalueweight <-median(Dataset$weight.kg.,na.rm=TRUE)
```

```
medianvalueweight
```

```
Dataset[is.na(Dataset$weight.kg.), "weight.kg."] <-medianvalueweight
```

```
Dataset
```

125	medianvalueweight <-median(Dataset\$weight.kg.,na.rm=TRUE)
126	medianvalueweight
127	Dataset[is.na(Dataset\$weight.kg.), "weight.kg."] <-medianvalueweight
128	Dataset

125:1 (Top Level) ⚙

Console	Terminal ×	Background Jobs ×
R 4.2.2 . ~/		
78	78	28 63.0 2 1 1 0 1 TRUE
79	79	25 58.0 1 2 3 0 1 FALSE
80	80	24 57.0 2 2 2 0 0 FALSE
> medianvalueweight <-median(Dataset\$weight.kg.,na.rm=TRUE)		
> medianvalueweight		
[1] 63.5		
> Dataset[is.na(Dataset\$weight.kg.), "weight.kg."] <-medianvalueweight		
> Dataset		
	id	Age weight.kg. Delivery_number Delivery_time Blood Heart Caesarian missing_value
1	1	22 57.7 1 0 1 0 0 FALSE
2	2	26 63.0 2 0 2 0 1 FALSE
3	3	26 62.0 2 1 2 0 0 FALSE
4	4	28 65.0 1 0 1 0 0 FALSE
5	5	22 58.0 2 0 2 0 1 FALSE
6	6	26 63.0 1 1 3 0 0 FALSE
7	7	27 64.0 2 0 2 0 0 FALSE
8	8	32 70.0 3 0 2 0 1 FALSE
9	9	28 63.5 2 0 <NA> 0 0 TRUE
10	10	27 64.5 1 1 2 0 1 FALSE
11	11	36 75.0 1 0 2 0 0 FALSE
12	12	33 70.0 1 1 3 0 1 FALSE
13	13	23 58.0 1 1 2 0 0 FALSE
14	14	20 55.0 1 0 2 1 0 FALSE
15	15	29 65.0 1 NA <NA> 1 1 TRUE
16	16	25 61.5 1 2 3 0 0 FALSE
17	17	25 61.5 1 0 2 0 0 FALSE
18	18	20 55.5 1 2 1 0 1 FALSE
19	19	37 76.0 3 0 2 1 1 FALSE
20	20	24 56.6 1 2 3 1 1 FALSE
21	21	26 62.0 1 1 2 0 0 FALSE
22	22	33 75.0 2 0 3 1 1 FALSE
23	23	25 62.0 1 1 1 0 0 FALSE
24	24	27 65.0 NA NA 3 1 1 TRUE
25	25	20 55.0 1 0 1 1 1 FALSE


```
medianvalueDelivery_number <-median(Dataset$Delivery_number,na.rm=TRUE)
```

```
medianvalueDelivery_number
```

```
Dataset[is.na(Dataset$Delivery_number), "Delivery_number"] <-medianvalueDelivery_number
```

```
Dataset
```

```
130 medianvalueDelivery_number <-median(Dataset$Delivery_number,na.rm=TRUE)
131 medianvalueDelivery_number
132 Dataset[is.na(Dataset$Delivery_number), "Delivery_number"] <-medianvalueDelivery_number
133 Dataset
```

130:1 (Top Level) ↕

Console Terminal × Background Jobs ×

R 4.2.2 . ~/

```
80 80 24 57.0 2 2 2 0 0 FALSE
> medianvalueDelivery_number <-median(Dataset$Delivery_number,na.rm=TRUE)
> medianvalueDelivery_number
[1] 1.5
> Dataset[is.na(Dataset$Delivery_number), "Delivery_number"] <-medianvalueDelivery_number
> Dataset
  id Age weight.kg Delivery_number Delivery_time Blood Heart Caesarian missing_value
1  1  22   57.7         1.0           0      1      0      0      FALSE
2  2  26   63.0         2.0           0      2      0      1      FALSE
3  3  26   62.0         2.0           1      2      0      0      FALSE
4  4  28   65.0         1.0           0      1      0      0      FALSE
5  5  22   58.0         2.0           0      2      0      1      FALSE
6  6  26   63.0         1.0           1      3      0      0      FALSE
7  7  27   64.0         2.0           0      2      0      0      FALSE
8  8  32   70.0         3.0           0      2      0      1      FALSE
9  9  28   63.5         2.0           0 <NA>      0      0      TRUE
10 10 27   64.5         1.0           1      2      0      1      FALSE
11 11 36   75.0         1.0           0      2      0      0      FALSE
12 12 33   70.0         1.0           1      3      0      1      FALSE
13 13 23   58.0         1.0           1      2      0      0      FALSE
14 14 20   55.0         1.0           0      2      1      0      FALSE
15 15 29   65.0         1.0           NA <NA>      1      1      TRUE
16 16 25   61.5         1.0           2      3      0      0      FALSE
17 17 25   61.5         1.0           0      2      0      0      FALSE
18 18 20   55.5         1.0           2      1      0      1      FALSE
19 19 37   76.0         3.0           0      2      1      1      FALSE
20 20 24   56.6         1.0           2      3      1      1      FALSE
21 21 26   62.0         1.0           1      2      0      0      FALSE
22 22 33   75.0         2.0           0      3      1      1      FALSE
23 23 25   62.0         1.0           1      1      0      0      FALSE
24 24 27   65.0         1.5           NA      3      1      1      TRUE
25 25 20   55.0         1.0           0      1      1      1      FALSE
26 26 18   49.0         1.5           0      2      0      0      TRUE
27 27 18   50.0         1.0           NA      1      1      1      TRUE
```

```
medianvalueDelivery_time <-median(Dataset$Delivery_time,na.rm=TRUE)
```

```
medianvalueDelivery_time
```

```
Dataset[is.na(Dataset$Delivery_time), "Delivery_time"] <-medianvalueDelivery_time
```

```
Dataset
```

```
135 medianvalueDelivery_time <-median(Dataset$Delivery_time,na.rm=TRUE)
136 medianvalueDelivery_time
137 Dataset[is.na(Dataset$Delivery_time), "Delivery_time"] <-medianvalueDelivery_time
138 Dataset
```

135:1 (Top Level) ▾

Console Terminal x Background Jobs x

R 4.2.2 ~ /

```
> medianvalueDelivery_time <-median(Dataset$Delivery_time,na.rm=TRUE)
> medianvalueDelivery_time
[1] 0
> Dataset[is.na(Dataset$Delivery_time), "Delivery_time"] <-medianvalueDelivery_time
> Dataset
```

	id	Age	weight.kg.	Delivery_number	Delivery_time	Blood	Heart	Caesarian	missing_value
1	1	22	57.7	1.0	0	1	0	0	FALSE
2	2	26	63.0	2.0	0	2	0	1	FALSE
3	3	26	62.0	2.0	1	2	0	0	FALSE
4	4	28	65.0	1.0	0	1	0	0	FALSE
5	5	22	58.0	2.0	0	2	0	1	FALSE
6	6	26	63.0	1.0	1	3	0	0	FALSE
7	7	27	64.0	2.0	0	2	0	0	FALSE
8	8	32	70.0	3.0	0	2	0	1	FALSE
9	9	28	63.5	2.0	0	<NA>	0	0	TRUE
10	10	27	64.5	1.0	1	2	0	1	FALSE
11	11	36	75.0	1.0	0	2	0	0	FALSE
12	12	33	70.0	1.0	1	3	0	1	FALSE
13	13	23	58.0	1.0	1	2	0	0	FALSE
14	14	20	55.0	1.0	0	2	1	0	FALSE
15	15	29	65.0	1.0	0	<NA>	1	1	TRUE
16	16	25	61.5	1.0	2	3	0	0	FALSE
17	17	25	61.5	1.0	0	2	0	0	FALSE
18	18	20	55.5	1.0	2	1	0	1	FALSE
19	19	37	76.0	3.0	0	2	1	1	FALSE
20	20	24	56.6	1.0	2	3	1	1	FALSE
21	21	26	62.0	1.0	1	2	0	0	FALSE
22	22	33	75.0	2.0	0	3	1	1	FALSE
23	23	25	62.0	1.0	1	1	0	0	FALSE
24	24	27	65.0	1.5	0	3	1	1	TRUE
25	25	20	55.0	1.0	0	1	1	1	FALSE
26	26	18	49.0	1.5	0	2	0	0	TRUE
27	27	18	50.0	1.0	0	1	1	1	TRUE

```
medianvalueCaesarian <-median(Dataset$Caesarian,na.rm=TRUE)
```

```
medianvalueCaesarian
```

```
Dataset[is.na(Dataset$Caesarian), "Caesarian"] <-medianvalueCaesarian
```

```
Dataset
```

```

142 medianvalueCaesarian <-median(Dataset$Caesarian,na.rm=TRUE)
143 medianvalueCaesarian
144 Dataset[is.na(Dataset$Caesarian), "Caesarian"] <-medianvalueCaesarian
145 Dataset
146

```

142:1 (Top Level) ↕

Console Terminal × Background Jobs ×

R 4.2.2 · D:/Semester/10) Spring 2023/Introduction To Data Science/Project/ ↗

```
> medianvalueCaesarian <-median(Dataset$Caesarian,na.rm=TRUE)
```

```
> medianvalueCaesarian
```

```
[1] 1
```

```
> Dataset[is.na(Dataset$Caesarian), "Caesarian"] <-medianvalueCaesarian
```

```
> Dataset
```

	id	Age	weight.kg.	Delivery_number	Delivery_time	Blood	Heart	Caesarian	missing_value
1	1	22	57.7	1.0	0	1	0	0	FALSE
2	2	26	63.0	2.0	0	2	0	1	FALSE
3	3	26	62.0	2.0	1	2	0	0	FALSE
4	4	28	65.0	1.0	0	1	0	0	FALSE
5	5	22	58.0	2.0	0	2	0	1	FALSE
6	6	26	63.0	1.0	1	3	0	0	FALSE
7	7	27	64.0	2.0	0	2	0	0	FALSE
8	8	32	70.0	3.0	0	2	0	1	FALSE
9	9	28	63.5	2.0	0	<NA>	0	0	TRUE
10	10	27	64.5	1.0	1	2	0	1	FALSE
11	11	36	75.0	1.0	0	2	0	0	FALSE
12	12	33	70.0	1.0	1	3	0	1	FALSE
13	13	23	58.0	1.0	1	2	0	0	FALSE
14	14	20	55.0	1.0	0	2	1	0	FALSE
15	15	29	65.0	1.0	0	<NA>	1	1	TRUE
16	16	25	61.5	1.0	2	3	0	0	FALSE
17	17	25	61.5	1.0	0	2	0	0	FALSE
18	18	20	55.5	1.0	2	1	0	1	FALSE
19	19	37	76.0	3.0	0	2	1	1	FALSE
20	20	24	56.6	1.0	2	3	1	1	FALSE
21	21	26	62.0	1.0	1	2	0	0	FALSE
22	22	33	75.0	2.0	0	3	1	1	FALSE
23	23	25	62.0	1.0	1	1	0	0	FALSE
24	24	27	65.0	1.5	0	3	1	1	TRUE
25	25	20	55.0	1.0	0	1	1	1	FALSE
26	26	18	49.0	1.5	0	2	0	0	TRUE
27	27	18	50.0	1.0	0	1	1	1	TRUE
28	28	30	68.0	1.0	0	2	0	0	FALSE
29	29	32	73.0	1.0	0	1	1	1	FALSE
30	30	26	62.5	2.0	1	2	1	0	FALSE
31	31	25	58.0	1.0	0	3	0	0	FALSE
32	32	40	82.0	1.0	0	2	1	1	FALSE
33	33	32	68.0	2.0	0	1	1	1	FALSE
34	34	27	63.0	2.0	0	2	1	1	FALSE
35	35	26	59.0	2.0	2	2	0	1	FALSE
36	36	28	66.0	3.0	0	1	0	1	FALSE
37	37	33	75.0	1.0	1	2	0	0	FALSE
38	38	31	69.0	2.0	2	2	0	0	FALSE
39	39	31	63.0	1.0	0	2	0	0	FALSE
40	40	26	59.0	1.0	2	3	1	1	FALSE
41	41	27	63.0	1.0	0	1	1	1	FALSE

42	42	19	51.0	1.0	0	2	0	1	FALSE
43	43	36	73.0	1.0	1	1	0	1	FALSE
44	44	22	57.0	1.0	0	2	0	1	FALSE
45	45	36	72.5	4.0	0	1	1	1	FALSE
46	46	28	62.5	3.0	0	2	1	1	FALSE
47	47	26	63.5	1.0	0	2	0	0	TRUE
48	48	32	67.5	2.0	0	1	1	1	FALSE
49	49	26	62.5	2.0	2	2	0	0	FALSE
50	50	28	63.5	2.0	0	3	1	1	TRUE
51	51	33	68.5	3.0	2	2	1	0	FALSE
52	52	21	53.0	2.0	1	3	1	1	FALSE
53	53	30	68.0	3.0	2	1	0	0	FALSE
54	54	35	74.0	1.0	1	3	0	0	FALSE
55	55	29	63.5	2.0	0	2	1	1	FALSE
56	56	25	59.0	2.0	0	2	0	0	FALSE
57	57	32	67.5	3.0	1	3	1	1	FALSE
58	58	95	110.0	1.0	0	3	0	1	FALSE
59	59	26	61.5	1.0	0	1	0	1	FALSE
60	60	30	67.5	2.0	1	1	1	1	TRUE
61	61	22	58.5	1.0	2	1	0	0	FALSE
62	62	28	63.5	1.0	0	2	0	1	TRUE
63	63	32	67.0	2.0	0	3	0	1	FALSE
64	64	32	67.0	2.0	0	2	1	1	FALSE
65	65	31	66.0	1.0	2	1	1	0	FALSE
66	66	35	72.0	2.0	0	2	0	1	FALSE
67	67	28	62.5	3.0	0	2	0	1	FALSE
68	68	29	64.5	2.0	0	2	1	0	FALSE
69	69	25	62.0	1.0	0	3	0	1	FALSE
70	70	27	61.0	2.0	2	3	0	0	FALSE
71	71	90	105.0	1.0	0	3	0	1	FALSE
72	72	29	65.0	1.0	2	<NA>	1	1	TRUE
73	73	28	64.0	2.0	0	2	0	0	FALSE
74	74	32	69.0	3.0	0	2	1	0	FALSE
75	75	38	75.0	3.0	2	1	1	1	FALSE
76	76	27	62.5	2.0	1	2	0	0	FALSE
77	77	33	66.0	4.0	0	2	0	1	TRUE
78	78	28	63.0	2.0	1	1	0	1	TRUE
79	79	25	58.0	1.0	2	3	0	1	FALSE
80	80	24	57.0	2.0	2	2	0	0	FALSE

```
##Median Blood
```

```
# Replace "NA" with NA in the 'Blood' column
```

```
Dataset$Blood[Dataset$Blood == "NA"] <- NA
```

```
# Convert the 'Blood' column to numeric
```

```
Dataset$Blood <- as.numeric(Dataset$Blood)
```

```
# Calculate the median excluding NAs
```

```
medianvalueBlood <- median(Dataset$Blood, na.rm = TRUE)
```

```
# Replace missing values in 'Blood' column with the median
```

```
Dataset$Blood[is.na(Dataset$Blood)] <- medianvalueBlood
```

```
Dataset
```



```

> # Replace "NA" with NA in the 'Blood' column
> Dataset$Blood[Dataset$Blood == "NA"] <- NA
> # Convert the 'Blood' column to numeric
> Dataset$Blood <- as.numeric(Dataset$Blood)
> # Calculate the median excluding NAs
> medianvalueBlood <- median(Dataset$Blood, na.rm = TRUE)
> # Replace missing values in 'Blood' column with the median
> Dataset$Blood[is.na(Dataset$Blood)] <- medianvalueBlood
> Dataset

```

	id	Age	weight.kg.	Delivery_number	Delivery_time	Blood	Heart	Caesarian	missing_value
1	1	22	57.7	1.0	0	1	0	0	FALSE
2	2	26	63.0	2.0	0	2	0	1	FALSE
3	3	26	62.0	2.0	1	2	0	0	FALSE
4	4	28	65.0	1.0	0	1	0	0	FALSE
5	5	22	58.0	2.0	0	2	0	1	FALSE
6	6	26	63.0	1.0	1	3	0	0	FALSE
7	7	27	64.0	2.0	0	2	0	0	FALSE
8	8	32	70.0	3.0	0	2	0	1	FALSE
9	9	28	63.5	2.0	0	2	0	0	TRUE
10	10	27	64.5	1.0	1	2	0	1	FALSE
11	11	36	75.0	1.0	0	2	0	0	FALSE
12	12	33	70.0	1.0	1	3	0	1	FALSE
13	13	23	58.0	1.0	1	2	0	0	FALSE
14	14	20	55.0	1.0	0	2	1	0	FALSE
15	15	29	65.0	1.0	0	2	1	1	TRUE
16	16	25	61.5	1.0	2	3	0	0	FALSE
17	17	25	61.5	1.0	0	2	0	0	FALSE
18	18	20	55.5	1.0	2	1	0	1	FALSE
19	19	37	76.0	3.0	0	2	1	1	FALSE
20	20	24	56.6	1.0	2	3	1	1	FALSE
21	21	26	62.0	1.0	1	2	0	0	FALSE
22	22	33	75.0	2.0	0	3	1	1	FALSE
23	23	25	62.0	1.0	1	1	0	0	FALSE
24	24	27	65.0	1.5	0	3	1	1	TRUE
25	25	20	55.0	1.0	0	1	1	1	FALSE
26	26	18	49.0	1.5	0	2	0	0	TRUE
27	27	18	50.0	1.0	0	1	1	1	TRUE
28	28	30	68.0	1.0	0	2	0	0	FALSE
29	29	32	73.0	1.0	0	1	1	1	FALSE
30	30	26	62.5	2.0	1	2	1	0	FALSE
31	31	25	58.0	1.0	0	3	0	0	FALSE
32	32	40	82.0	1.0	0	2	1	1	FALSE
33	33	32	68.0	2.0	0	1	1	1	FALSE
34	34	27	63.0	2.0	0	2	1	1	FALSE
35	35	26	59.0	2.0	2	2	0	1	FALSE
36	36	28	66.0	3.0	0	1	0	1	FALSE
37	37	33	75.0	1.0	1	2	0	0	FALSE
38	38	31	69.0	2.0	2	2	0	0	FALSE
39	39	31	63.0	1.0	0	2	0	0	FALSE
40	40	26	59.0	1.0	2	3	1	1	FALSE
41	41	27	63.0	1.0	0	1	1	1	FALSE
42	42	19	51.0	1.0	0	2	0	1	FALSE
43	43	36	73.0	1.0	1	1	0	1	FALSE
44	44	22	57.0	1.0	0	2	0	1	FALSE
45	45	36	72.5	4.0	0	1	1	1	FALSE
46	46	28	62.5	3.0	0	2	1	1	FALSE
47	47	26	63.5	1.0	0	2	0	0	TRUE
48	48	32	67.5	2.0	0	1	1	1	FALSE
49	49	26	62.5	2.0	2	2	0	0	FALSE
50	50	28	63.5	2.0	0	3	1	1	TRUE
51	51	33	68.5	3.0	2	2	1	0	FALSE
52	52	21	53.0	2.0	1	3	1	1	FALSE
53	53	30	68.0	3.0	2	1	0	0	FALSE
54	54	35	74.0	1.0	1	3	0	0	FALSE
55	55	29	63.5	2.0	0	2	1	1	FALSE
56	56	25	59.0	2.0	0	2	0	0	FALSE
57	57	32	67.5	3.0	1	3	1	1	FALSE
58	58	95	110.0	1.0	0	3	0	1	FALSE
59	59	26	61.5	1.0	0	1	0	1	FALSE
60	60	30	67.5	2.0	1	1	1	1	TRUE
61	61	22	58.5	1.0	2	1	0	0	FALSE
62	62	28	63.5	1.0	0	2	0	1	TRUE
63	63	32	67.0	2.0	0	3	0	1	FALSE
64	64	32	67.0	2.0	0	2	1	1	FALSE
65	65	31	66.0	1.0	2	1	1	0	FALSE
66	66	35	72.0	2.0	0	2	0	1	FALSE
67	67	28	62.5	3.0	0	2	0	1	FALSE
68	68	29	64.5	2.0	0	2	1	0	FALSE
69	69	25	62.0	1.0	0	3	0	1	FALSE
70	70	27	61.0	2.0	2	3	0	0	FALSE
71	71	90	105.0	1.0	0	3	0	1	FALSE
72	72	29	65.0	1.0	2	2	1	1	TRUE
73	73	28	64.0	2.0	0	2	0	0	FALSE
74	74	32	69.0	3.0	0	2	1	0	FALSE
75	75	38	75.0	3.0	2	1	1	1	FALSE
76	76	27	62.5	2.0	1	2	0	0	FALSE
77	77	33	66.0	4.0	0	2	0	1	TRUE
78	78	28	63.0	2.0	1	1	0	1	TRUE
79	79	25	58.0	1.0	2	3	0	1	FALSE
80	80	24	57.0	2.0	2	2	0	0	FALSE

III. Recover missing values with the mode value.

```
modeAge=names(sort(table(Dataset$Age)))[1]
```

```
modeAge
```

```
Dataset[is.na(Dataset$Age), "Age"] <-modeAge
```

```
Dataset
```

```
145 modeAge=names(sort(table(Dataset$Age)))[1]
146 modeAge
147 Dataset[is.na(Dataset$Age), "Age"] <-modeAge
148 Dataset
149
150
151
```

145:1 (Top Level) ↕

Console Terminal × Background Jobs ×

R 4.2.2 · ~/

```
> modeAge
[1] "19"
> Dataset[is.na(Dataset$Age), "Age"] <-modeAge
> Dataset
```

	id	Age	weight.kg.	Delivery_number	Delivery_time	Blood	Heart	Caesarian	missing_value
1	1	22	57.7	1	0	1	0	0	FALSE
2	2	26	63.0	2	0	2	0	1	FALSE
3	3	26	62.0	2	1	2	0	0	FALSE
4	4	28	65.0	1	0	1	0	0	FALSE
5	5	22	58.0	2	0	2	0	1	FALSE
6	6	26	63.0	1	1	3	0	0	FALSE
7	7	27	64.0	2	0	2	0	0	FALSE
8	8	32	70.0	3	0	2	0	1	FALSE
9	9	28	63.5	2	0	<NA>	0	0	TRUE
10	10	27	64.5	1	1	2	0	1	FALSE
11	11	36	75.0	1	0	2	0	0	FALSE
12	12	33	70.0	1	1	3	0	1	FALSE
13	13	23	58.0	1	1	2	0	0	FALSE
14	14	20	55.0	1	0	2	1	0	FALSE
15	15	29	65.0	1	NA	<NA>	1	1	TRUE
16	16	25	61.5	1	2	3	0	0	FALSE
17	17	25	61.5	1	0	2	0	0	FALSE
18	18	20	55.5	1	2	1	0	1	FALSE
19	19	37	76.0	3	0	2	1	1	FALSE
20	20	24	56.6	1	2	3	1	1	FALSE
21	21	26	62.0	1	1	2	0	0	FALSE
22	22	33	75.0	2	0	3	1	1	FALSE
23	23	25	62.0	1	1	1	0	0	FALSE
24	24	27	65.0	NA	NA	3	1	1	TRUE
25	25	20	55.0	1	0	1	1	1	FALSE
26	26	18	49.0	NA	0	2	0	0	TRUE
27	27	18	50.0	1	NA	1	1	1	TRUE
28	28	30	68.0	1	0	2	0	0	FALSE
29	29	32	73.0	1	0	1	1	1	FALSE

```
modeweight=names(sort(table(Dataset$weight.kg.)))[1]
```

```
modeweight
```

```
Dataset[is.na(Dataset$weight.kg.), "weight.kg."] <-modeweight
```

```
Dataset
```

152 modeweight=names(sort(table(Dataset\$weight.kg.)))[1]
153 modeweight
154 Dataset[is.na(Dataset\$weight.kg.), "weight.kg."] <-modeweight
155 Dataset

152:1 (Top Level) ↕

Console Terminal × Background Jobs ×

R 4.2.2 · ~/ ↗

80 80 24 57.0 2 2 2 0 0 FALSE
> modeweight=names(sort(table(Dataset\$weight.kg.)))[1]
> modeweight
[1] "49"
> Dataset[is.na(Dataset\$weight.kg.), "weight.kg."] <-modeweight
> Dataset

	id	Age	weight.kg.	Delivery_number	Delivery_time	Blood	Heart	Caesarian	missing_value
1	1	22	57.7	1	0	1	0	0	FALSE
2	2	26	63	2	0	2	0	1	FALSE
3	3	26	62	2	1	2	0	0	FALSE
4	4	28	65	1	0	1	0	0	FALSE
5	5	22	58	2	0	2	0	1	FALSE
6	6	26	63	1	1	3	0	0	FALSE
7	7	27	64	2	0	2	0	0	FALSE
8	8	32	70	3	0	2	0	1	FALSE
9	9	28	63.5	2	0	<NA>	0	0	TRUE
10	10	27	64.5	1	1	2	0	1	FALSE
11	11	36	75	1	0	2	0	0	FALSE
12	12	33	70	1	1	3	0	1	FALSE
13	13	23	58	1	1	2	0	0	FALSE
14	14	20	55	1	0	2	1	0	FALSE
15	15	29	65	1	NA	<NA>	1	1	TRUE
16	16	25	61.5	1	2	3	0	0	FALSE
17	17	25	61.5	1	0	2	0	0	FALSE
18	18	20	55.5	1	2	1	0	1	FALSE
19	19	37	76	3	0	2	1	1	FALSE
20	20	24	56.6	1	2	3	1	1	FALSE
21	21	26	62	1	1	2	0	0	FALSE
22	22	33	75	2	0	3	1	1	FALSE
23	23	25	62	1	1	1	0	0	FALSE
24	24	27	65	NA	NA	3	1	1	TRUE
25	25	20	55	1	0	1	1	1	FALSE
26	26	18	49	NA	0	2	0	0	TRUE
27	27	18	50	1	NA	1	1	1	TRUE

```
157 modeDelivery_number=names(sort(table(Dataset$Delivery_number)))[1]
158 modeDelivery_number
159 Dataset[is.na(Dataset$Delivery_number), "Delivery_number"] <-modeDelivery_number
160 Dataset
```

157:1 (Top Level) ⚡

Console Terminal Background Jobs

R 4.2.2, ~/

```
80 80 24 57 2 2 2 0 0 FALSE
> modeDelivery_number=names(sort(table(Dataset$Delivery_number)))[1]
> modeDelivery_number
[1] "4"
> Dataset[is.na(Dataset$Delivery_number), "Delivery_number"] <-modeDelivery_number
> Dataset
```

	id	Age	weight.kg.	Delivery_number	Delivery_time	Blood	Heart	Caesarian	missing_value
1	1	22	57.7	1	0	1	0	0	FALSE
2	2	26	63	2	0	2	0	1	FALSE
3	3	26	62	2	1	2	0	0	FALSE
4	4	28	65	1	0	1	0	0	FALSE
5	5	22	58	2	0	2	0	1	FALSE
6	6	26	63	1	1	3	0	0	FALSE
7	7	27	64	2	0	2	0	0	FALSE
8	8	32	70	3	0	2	0	1	FALSE
9	9	28	63.5	2	0	<NA>	0	0	TRUE
10	10	27	64.5	1	1	2	0	1	FALSE
11	11	36	75	1	0	2	0	0	FALSE
12	12	33	70	1	1	3	0	1	FALSE
13	13	23	58	1	1	2	0	0	FALSE
14	14	20	55	1	0	2	1	0	FALSE
15	15	29	65	1	NA	<NA>	1	1	TRUE
16	16	25	61.5	1	2	3	0	0	FALSE
17	17	25	61.5	1	0	2	0	0	FALSE
18	18	20	55.5	1	2	1	0	1	FALSE
19	19	37	76	3	0	2	1	1	FALSE
20	20	24	56.6	1	2	3	1	1	FALSE
21	21	26	62	1	1	2	0	0	FALSE
22	22	33	75	2	0	3	1	1	FALSE
23	23	25	62	1	1	1	0	0	FALSE
24	24	27	65	4	NA	3	1	1	TRUE
25	25	20	55	1	0	1	1	1	FALSE
26	26	18	49	4	0	2	0	0	TRUE
27	27	18	50	1	NA	1	1	1	TRUE


```
modeDelivery_time=names(sort(table(Dataset$Delivery_time)))[1]
```

```
modeDelivery_time
```

```
Dataset[is.na(Dataset$Delivery_time), "Delivery_time"] <-modeDelivery_time
```

```
Dataset
```

```
162 modeDelivery_time=names(sort(table(Dataset$Delivery_time)))[1]
163 modeDelivery_time
164 Dataset[is.na(Dataset$Delivery_time), "Delivery_time"] <-modeDelivery_time
165 Dataset
166
167
```

162:1 (Top Level) ↕

Console Terminal Background Jobs

R 4.2.2 ~ /

```
> modeDelivery_time=names(sort(table(Dataset$Delivery_time)))[1]
> modeDelivery_time
[1] "1"
> Dataset[is.na(Dataset$Delivery_time), "Delivery_time"] <-modeDelivery_time
> Dataset
```

	id	Age	weight.kg.	Delivery_number	Delivery_time	Blood	Heart	Caesarian	missing_value
1	1	22	57.7	1	0	1	0	0	FALSE
2	2	26	63	2	0	2	0	1	FALSE
3	3	26	62	2	1	2	0	0	FALSE
4	4	28	65	1	0	1	0	0	FALSE
5	5	22	58	2	0	2	0	1	FALSE
6	6	26	63	1	1	3	0	0	FALSE
7	7	27	64	2	0	2	0	0	FALSE
8	8	32	70	3	0	2	0	1	FALSE
9	9	28	63.5	2	0	<NA>	0	0	TRUE
10	10	27	64.5	1	1	2	0	1	FALSE
11	11	36	75	1	0	2	0	0	FALSE
12	12	33	70	1	1	3	0	1	FALSE
13	13	23	58	1	1	2	0	0	FALSE
14	14	20	55	1	0	2	1	0	FALSE
15	15	29	65	1	1	<NA>	1	1	TRUE
16	16	25	61.5	1	2	3	0	0	FALSE
17	17	25	61.5	1	0	2	0	0	FALSE
18	18	20	55.5	1	2	1	0	1	FALSE
19	19	37	76	3	0	2	1	1	FALSE
20	20	24	56.6	1	2	3	1	1	FALSE
21	21	26	62	1	1	2	0	0	FALSE
22	22	33	75	2	0	3	1	1	FALSE
23	23	25	62	1	1	1	0	0	FALSE
24	24	27	65	4	1	3	1	1	TRUE
25	25	20	55	1	0	1	1	1	FALSE
26	26	18	49	4	0	2	0	0	TRUE
27	27	18	50	1	1	1	1	1	TRUE
28	28	30	68	1	0	2	0	0	FALSE

```
modeCaesarian=names(sort(table(Dataset$Caesarian)))[1]
```

```
modeCaesarian
```

```
Dataset[is.na(Dataset$Caesarian), "Caesarian"] <-modeCaesarian
```

```
Dataset
```

```

167 modeCaesarian=names(sort(table(Dataset$Caesarian)))[1]
168 modeCaesarian
169 Dataset[is.na(Dataset$Caesarian), "Caesarian"] <-modeCaesarian
170 Dataset
171
172

```

167:1 (Top Level) ↕

Console Terminal × Background Jobs ×

R 4.2.2 · ~/

```
80 80 24 57 2 2 2 0 0 FALSE
```

```
> modeCaesarian=names(sort(table(Dataset$Caesarian)))[1]
```

```
> modeCaesarian
```

```
[1] "0"
```

```
> Dataset[is.na(Dataset$Caesarian), "Caesarian"] <-modeCaesarian
```

```
> Dataset
```

	id	Age	weight.kg.	Delivery_number	Delivery_time	Blood	Heart	Caesarian	missing_value
1	1	22	57.7	1	0	1	0	0	FALSE
2	2	26	63	2	0	2	0	1	FALSE
3	3	26	62	2	1	2	0	0	FALSE
4	4	28	65	1	0	1	0	0	FALSE
5	5	22	58	2	0	2	0	1	FALSE
6	6	26	63	1	1	3	0	0	FALSE
7	7	27	64	2	0	2	0	0	FALSE
8	8	32	70	3	0	2	0	1	FALSE
9	9	28	63.5	2	0	<NA>	0	0	TRUE
10	10	27	64.5	1	1	2	0	1	FALSE
11	11	36	75	1	0	2	0	0	FALSE
12	12	33	70	1	1	3	0	1	FALSE
13	13	23	58	1	1	2	0	0	FALSE
14	14	20	55	1	0	2	1	0	FALSE
15	15	29	65	1	1	<NA>	1	1	TRUE
16	16	25	61.5	1	2	3	0	0	FALSE
17	17	25	61.5	1	0	2	0	0	FALSE
18	18	20	55.5	1	2	1	0	1	FALSE
19	19	37	76	3	0	2	1	1	FALSE
20	20	24	56.6	1	2	3	1	1	FALSE
21	21	26	62	1	1	2	0	0	FALSE
22	22	33	75	2	0	3	1	1	FALSE
23	23	25	62	1	1	1	0	0	FALSE
24	24	27	65	4	1	3	1	1	TRUE
25	25	20	55	1	0	1	1	1	FALSE
26	26	18	49	4	0	2	0	0	TRUE
27	27	18	50	1	1	1	1	1	TRUE
28	28	30	68	1	0	2	0	0	FALSE
29	29	32	73	1	0	1	1	1	FALSE
30	30	26	62.5	2	1	2	1	0	FALSE
31	31	25	58	1	0	3	0	0	FALSE
32	32	40	82	1	0	2	1	1	FALSE
33	33	32	68	2	0	1	1	1	FALSE
34	34	27	63	2	0	2	1	1	FALSE
35	35	26	59	2	2	2	0	1	FALSE
36	36	28	66	3	0	1	0	1	FALSE
37	37	33	75	1	1	2	0	0	FALSE
38	38	31	69	2	2	2	0	0	FALSE
39	39	31	63	1	0	2	0	0	FALSE
40	40	26	59	1	2	3	1	1	FALSE
41	41	27	63	1	0	1	1	1	FALSE
42	42	19	51	1	0	2	0	1	FALSE
43	43	36	73	1	1	1	0	1	FALSE
44	44	22	57	1	0	2	0	1	FALSE
45	45	36	72.5	4	0	1	1	1	FALSE
46	46	28	62.5	3	0	2	1	1	FALSE
47	47	26	49	1	0	2	0	0	TRUE
48	48	32	67.5	2	0	1	1	1	FALSE
49	49	26	62.5	2	2	2	0	0	FALSE
50	50	19	49	2	0	3	1	1	TRUE

51	51	33	68.5	3	2	2	1	0	FALSE
52	52	21	53	2	1	3	1	1	FALSE
53	53	30	68	3	2	1	0	0	FALSE
54	54	35	74	1	1	3	0	0	FALSE
55	55	29	63.5	2	0	2	1	1	FALSE
56	56	25	59	2	0	2	0	0	FALSE
57	57	32	67.5	3	1	3	1	1	FALSE
58	58	95	110	1	0	3	0	1	FALSE
59	59	26	61.5	1	0	1	0	1	FALSE
60	60	30	67.5	2	1	1	1	0	TRUE
61	61	22	58.5	1	2	1	0	0	FALSE
62	62	19	49	1	0	2	0	1	TRUE
63	63	32	67	2	0	3	0	1	FALSE
64	64	32	67	2	0	2	1	1	FALSE
65	65	31	66	1	2	1	1	0	FALSE
66	66	35	72	2	0	2	0	1	FALSE
67	67	28	62.5	3	0	2	0	1	FALSE
68	68	29	64.5	2	0	2	1	0	FALSE
69	69	25	62	1	0	3	0	1	FALSE
70	70	27	61	2	2	3	0	0	FALSE
71	71	90	105	1	0	3	0	1	FALSE
72	72	29	65	1	2	<NA>	1	1	TRUE
73	73	28	64	2	0	2	0	0	FALSE
74	74	32	69	3	0	2	1	0	FALSE
75	75	38	75	3	2	1	1	1	FALSE
76	76	27	62.5	2	1	2	0	0	FALSE
77	77	33	66	4	0	2	0	0	TRUE
78	78	19	63	2	1	1	0	1	TRUE
79	79	25	58	1	2	3	0	1	FALSE
80	80	24	57	2	2	2	0	0	FALSE

#####Mode Blood

Replace "NA" with NA in the 'Blood' column

Dataset\$Blood[Dataset\$Blood == "NA"] <- NA

Convert the 'Blood' column to numeric

Dataset\$Blood <- as.numeric(Dataset\$Blood)

Function to calculate the mode

```
calculate_mode <- function(x) {
```

```
  tbl <- table(x)
```

```
  modes <- tbl[tbl == max(tbl)]
```

```
  as.numeric(names(modes))
```

```
}
```

Calculate the mode excluding NAs

modeBlood <- calculate_mode(Dataset\$Blood[!is.na(Dataset\$Blood)])

Replace missing values in 'Blood' column with the mode

Dataset\$Blood[is.na(Dataset\$Blood)] <- modeBlood

Dataset

```

> #####Mode Blood
> # Replace "NA" with NA in the 'Blood' column
> Dataset$Blood[Dataset$Blood == "NA"] <- NA
> # Convert the 'Blood' column to numeric
> Dataset$Blood <- as.numeric(Dataset$Blood)
> # Function to calculate the mode
> calculate_mode <- function(x) {
+   tbl <- table(x)
+   modes <- tbl[tbl == max(tbl)]
+   as.numeric(names(modes))
+ }
> # Calculate the mode excluding NAs
> modeBlood <- calculate_mode(Dataset$Blood[!is.na(Dataset$Blood)])
> # Replace missing values in 'Blood' column with the mode
> Dataset$Blood[is.na(Dataset$Blood)] <- modeBlood
> Dataset

```

	id	Age	weight.kg.	Delivery_number	Delivery_time	Blood	Heart	Caesarian	missing_value
1	1	22	57.7	1	0	1	0	0	FALSE
2	2	26	63	2	0	2	0	1	FALSE
3	3	26	62	2	1	2	0	0	FALSE
4	4	28	65	1	0	1	0	0	FALSE
5	5	22	58	2	0	2	0	1	FALSE
6	6	26	63	1	1	3	0	0	FALSE
7	7	27	64	2	0	2	0	0	FALSE
8	8	32	70	3	0	2	0	1	FALSE
9	9	28	63.5	2	0	2	0	0	TRUE
10	10	27	64.5	1	1	2	0	1	FALSE
11	11	36	75	1	0	2	0	0	FALSE
12	12	33	70	1	1	3	0	1	FALSE
13	13	23	58	1	1	2	0	0	FALSE
14	14	20	55	1	0	2	1	0	FALSE
15	15	29	65	1	1	2	1	1	TRUE
16	16	25	61.5	1	2	3	0	0	FALSE
17	17	25	61.5	1	0	2	0	0	FALSE
18	18	20	55.5	1	2	1	0	1	FALSE
19	19	37	76	3	0	2	1	1	FALSE
20	20	24	56.6	1	2	3	1	1	FALSE
21	21	26	62	1	1	2	0	0	FALSE
22	22	33	75	2	0	3	1	1	FALSE
23	23	25	62	1	1	1	0	0	FALSE
24	24	27	65	4	1	3	1	1	TRUE
25	25	20	55	1	0	1	1	1	FALSE
26	26	18	49	4	0	2	0	0	TRUE
27	27	18	50	1	1	1	1	1	TRUE
28	28	30	68	1	0	2	0	0	FALSE
29	29	32	73	1	0	1	1	1	FALSE
30	30	26	62.5	2	1	2	1	0	FALSE
31	31	25	58	1	0	3	0	0	FALSE
32	32	40	82	1	0	2	1	1	FALSE
33	33	32	68	2	0	1	1	1	FALSE
34	34	27	63	2	0	2	1	1	FALSE
35	35	26	59	2	2	2	0	1	FALSE
36	36	28	66	3	0	1	0	1	FALSE
37	37	33	75	1	1	2	0	0	FALSE
38	38	31	69	2	2	2	0	0	FALSE
39	39	31	63	1	0	2	0	0	FALSE
40	40	26	59	1	2	3	1	1	FALSE
41	41	27	63	1	0	1	1	1	FALSE
42	42	19	51	1	0	2	0	1	FALSE
43	43	36	73	1	1	1	0	1	FALSE
44	44	22	57	1	0	2	0	1	FALSE
45	45	36	72.5	4	0	1	1	1	FALSE
46	46	28	62.5	3	0	2	1	1	FALSE
47	47	26	49	1	0	2	0	0	TRUE
48	48	32	67.5	2	0	1	1	1	FALSE
49	49	26	62.5	2	2	2	0	0	FALSE
50	50	19	49	2	0	3	1	1	TRUE
51	51	33	68.5	3	2	2	1	0	FALSE
52	52	21	53	2	1	3	1	1	FALSE
53	53	30	68	3	2	1	0	0	FALSE
54	54	35	74	1	1	3	0	0	FALSE
55	55	29	63.5	2	0	2	1	1	FALSE
56	56	25	59	2	0	2	0	0	FALSE
57	57	32	67.5	3	1	3	1	1	FALSE
58	58	95	110	1	0	3	0	1	FALSE
59	59	26	61.5	1	0	1	0	1	FALSE
60	60	30	67.5	2	1	1	1	0	TRUE
61	61	22	58.5	1	2	1	0	0	FALSE
62	62	19	49	1	0	2	0	1	TRUE
63	63	32	67	2	0	3	0	1	FALSE
64	64	32	67	2	0	2	1	1	FALSE
65	65	31	66	1	2	1	1	0	FALSE
66	66	35	72	2	0	2	0	1	FALSE
67	67	28	62.5	3	0	2	0	1	FALSE
68	68	29	64.5	2	0	2	1	0	FALSE
69	69	25	62	1	0	3	0	1	FALSE
70	70	27	61	2	2	3	0	0	FALSE
71	71	90	105	1	0	3	0	1	FALSE
72	72	29	65	1	2	2	1	1	TRUE
73	73	28	64	2	0	2	0	0	FALSE
74	74	32	69	3	0	2	1	0	FALSE
75	75	38	75	3	2	1	1	1	FALSE
76	76	27	62.5	2	1	2	0	0	FALSE
77	77	33	66	4	0	2	0	0	TRUE
78	78	19	63	2	1	1	0	1	TRUE
79	79	25	58	1	2	3	0	1	FALSE
80	80	24	57	2	2	2	0	0	FALSE

IV. Remove the rows that have missing values from the data set.

```
remove_missingvalue<- na.omit(Dataset)
```

```
remove_missingvalue
```

90	remove_missingvalue<- na.omit(Dataset)
91	remove_missingvalue
92	

90:1	(Top Level) ⌵
------	---------------

Console	Terminal	Background Jobs
---------	----------	-----------------

R 4.2.2 . ~/

```
> remove_missingvalue<- na.omit(Dataset)
> remove_missingvalue
```

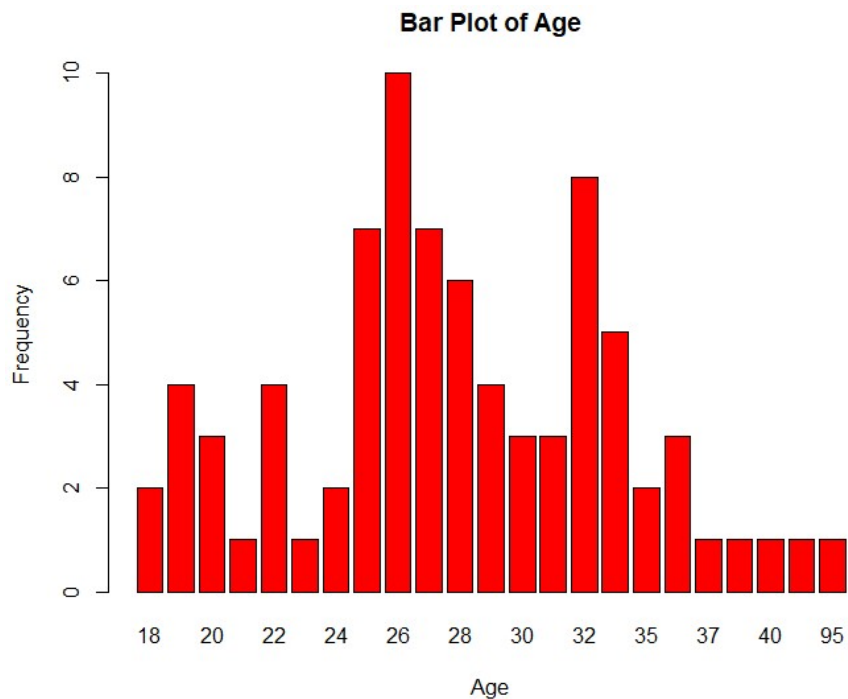
	id	Age	weight.kg.	Delivery_number	Delivery_time	Blood	Heart	Caesarian	missing_value
1	1	22	57.7	1	0	1	0	0	FALSE
2	2	26	63.0	2	0	2	0	1	FALSE
3	3	26	62.0	2	1	2	0	0	FALSE
4	4	28	65.0	1	0	1	0	0	FALSE
5	5	22	58.0	2	0	2	0	1	FALSE
6	6	26	63.0	1	1	3	0	0	FALSE
7	7	27	64.0	2	0	2	0	0	FALSE
8	8	32	70.0	3	0	2	0	1	FALSE
10	10	27	64.5	1	1	2	0	1	FALSE
11	11	36	75.0	1	0	2	0	0	FALSE
12	12	33	70.0	1	1	3	0	1	FALSE
13	13	23	58.0	1	1	2	0	0	FALSE
14	14	20	55.0	1	0	2	1	0	FALSE
16	16	25	61.5	1	2	3	0	0	FALSE
17	17	25	61.5	1	0	2	0	0	FALSE
18	18	20	55.5	1	2	1	0	1	FALSE
19	19	37	76.0	3	0	2	1	1	FALSE
20	20	24	56.6	1	2	3	1	1	FALSE
21	21	26	62.0	1	1	2	0	0	FALSE
22	22	33	75.0	2	0	3	1	1	FALSE
23	23	25	62.0	1	1	1	0	0	FALSE
25	25	20	55.0	1	0	1	1	1	FALSE
28	28	30	68.0	1	0	2	0	0	FALSE
29	29	32	73.0	1	0	1	1	1	FALSE
30	30	26	62.5	2	1	2	1	0	FALSE
31	31	25	58.0	1	0	3	0	0	FALSE
32	32	40	82.0	1	0	2	1	1	FALSE
33	33	32	68.0	2	0	1	1	1	FALSE
34	34	27	63.0	2	0	2	1	1	FALSE
35	35	26	59.0	2	2	2	0	1	FALSE
36	36	28	66.0	3	0	1	0	1	FALSE
37	37	33	75.0	1	1	2	0	0	FALSE
38	38	31	69.0	2	2	2	0	0	FALSE
39	39	31	63.0	1	0	2	0	0	FALSE
40	40	26	59.0	1	2	3	1	1	FALSE
41	41	27	63.0	1	0	1	1	1	FALSE
42	42	19	51.0	1	0	2	0	1	FALSE
43	43	36	73.0	1	1	1	0	1	FALSE
44	44	22	57.0	1	0	2	0	1	FALSE
45	45	36	72.5	4	0	1	1	1	FALSE
46	46	28	62.5	3	0	2	1	1	FALSE
48	48	32	67.5	2	0	1	1	1	FALSE
49	49	26	62.5	2	2	2	0	0	FALSE
51	51	33	68.5	3	2	2	1	0	FALSE
52	52	21	53.0	2	1	3	1	1	FALSE
53	53	30	68.0	3	2	1	0	0	FALSE
54	54	35	74.0	1	1	3	0	0	FALSE
55	55	29	63.5	2	0	2	1	1	FALSE
56	56	25	59.0	2	0	2	0	0	FALSE
57	57	32	67.5	3	1	3	1	1	FALSE
58	58	95	110.0	1	0	3	0	1	FALSE
59	59	26	61.5	1	0	1	0	1	FALSE
61	61	22	58.5	1	2	1	0	0	FALSE
63	63	32	67.0	2	0	3	0	1	FALSE
64	64	32	67.0	2	0	2	1	1	FALSE
65	65	31	66.0	1	2	1	1	0	FALSE
66	66	35	72.0	2	0	2	0	1	FALSE
67	67	28	62.5	3	0	2	0	1	FALSE
68	68	29	64.5	2	0	2	1	0	FALSE
69	69	25	62.0	1	0	3	0	1	FALSE
70	70	27	61.0	2	2	3	0	0	FALSE
71	71	90	105.0	1	0	3	0	1	FALSE
73	73	28	64.0	2	0	2	0	0	FALSE
74	74	32	69.0	3	0	2	1	0	FALSE
75	75	38	75.0	3	2	1	1	1	FALSE
76	76	27	62.5	2	1	2	0	0	FALSE
79	79	25	58.0	1	2	3	0	1	FALSE
80	80	24	57.0	2	2	2	0	0	FALSE

16) Data Visualization

Bar Plot:

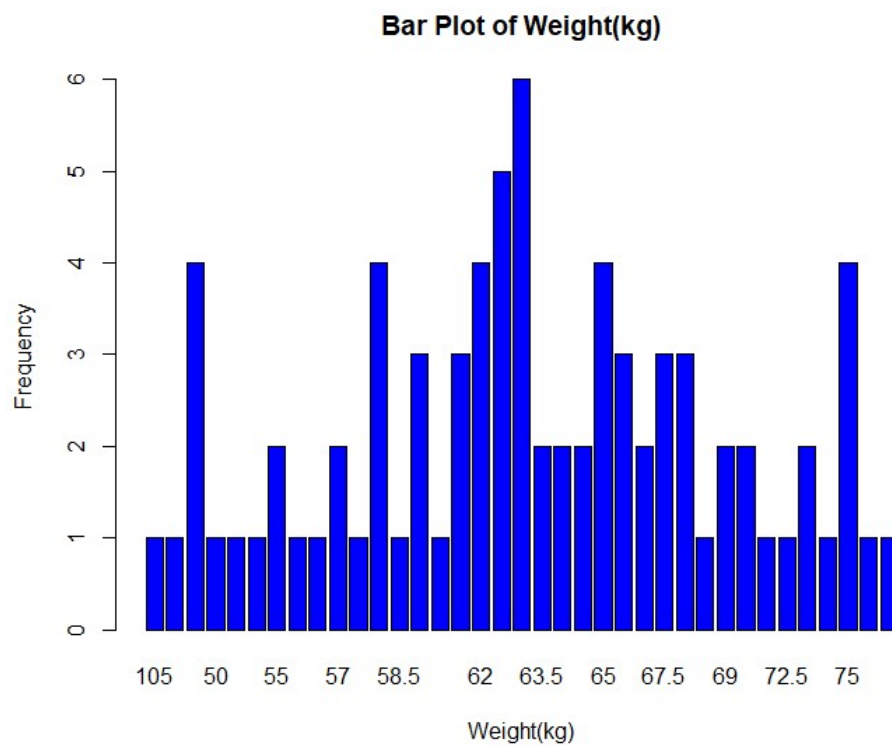
Bar Plot of Age:

```
barplot(table(Dataset$Age), main = "Bar Plot of Age", xlab = "Age", ylab = "Frequency", col = "red")
```



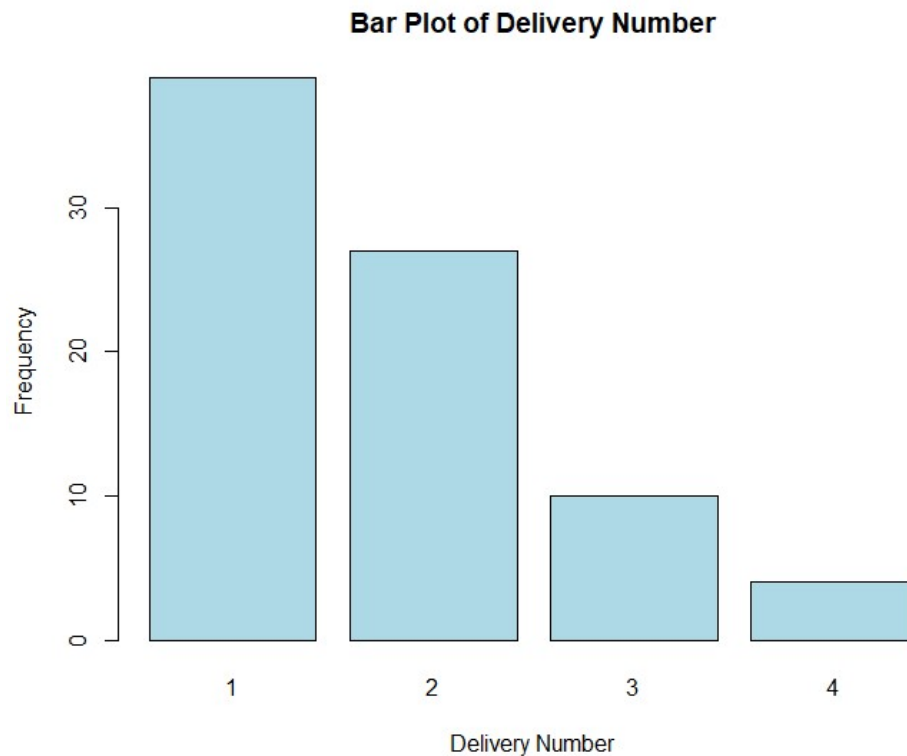
Bar Plot of Weight(kg):

```
barplot(table(Dataset$weight.kg.), main = "Bar Plot of Weight(kg)", xlab = "Weight(kg)", ylab = "Frequency", col = "blue")
```



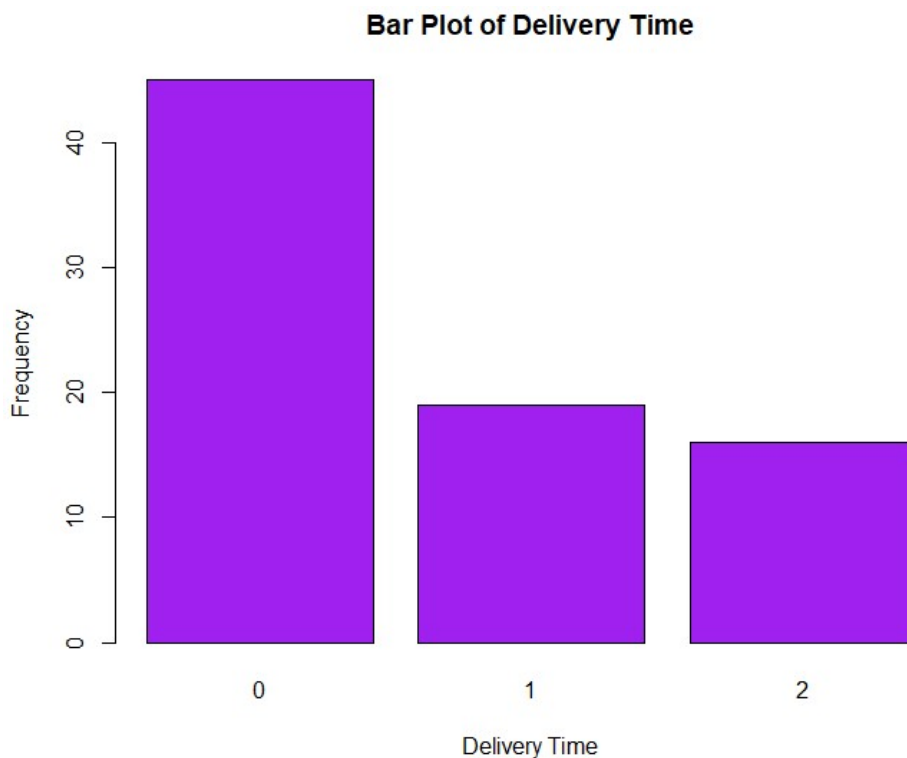
Bar Plot of Delivery Number:

```
barplot(table(Dataset$Delivery_number), main = "Bar Plot of Delivery Number", xlab = "Delivery  
Number", ylab = "Frequency", col = "lightblue")
```



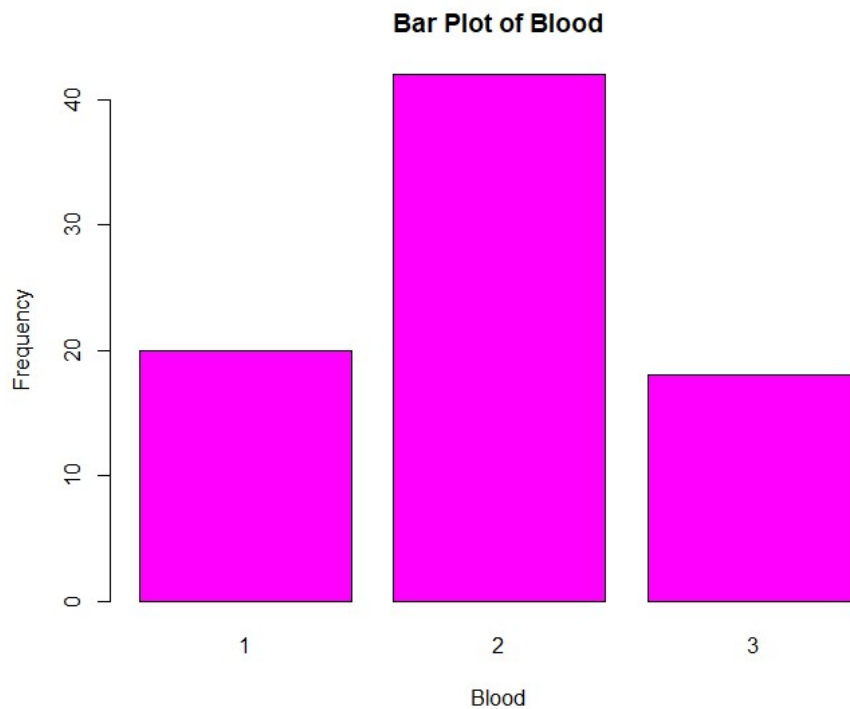
Bar Plot of Delivery Time:

```
barplot(table(Dataset$Delivery_time), main = "Bar Plot of Delivery Time", xlab = "Delivery Time",  
ylab = "Frequency", col = "purple")
```



Bar Plot of Blood:

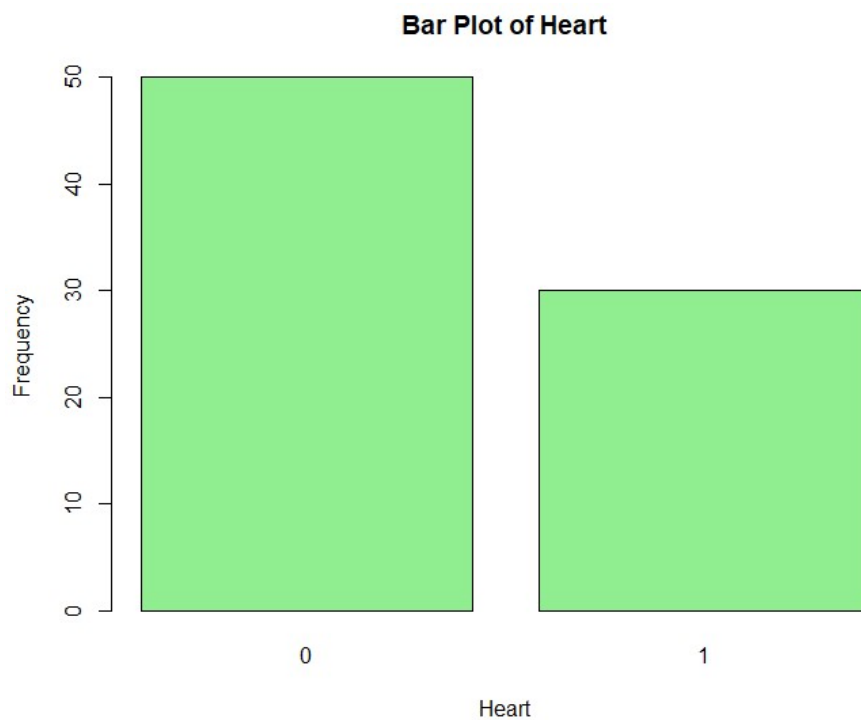
```
barplot(table(Dataset$Blood), main = "Bar Plot of Blood", xlab = "Blood", ylab = "Frequency", col = "magenta")
```



N.B: Annotated high as 1, normal as 2, and low as 3 from the Blood.

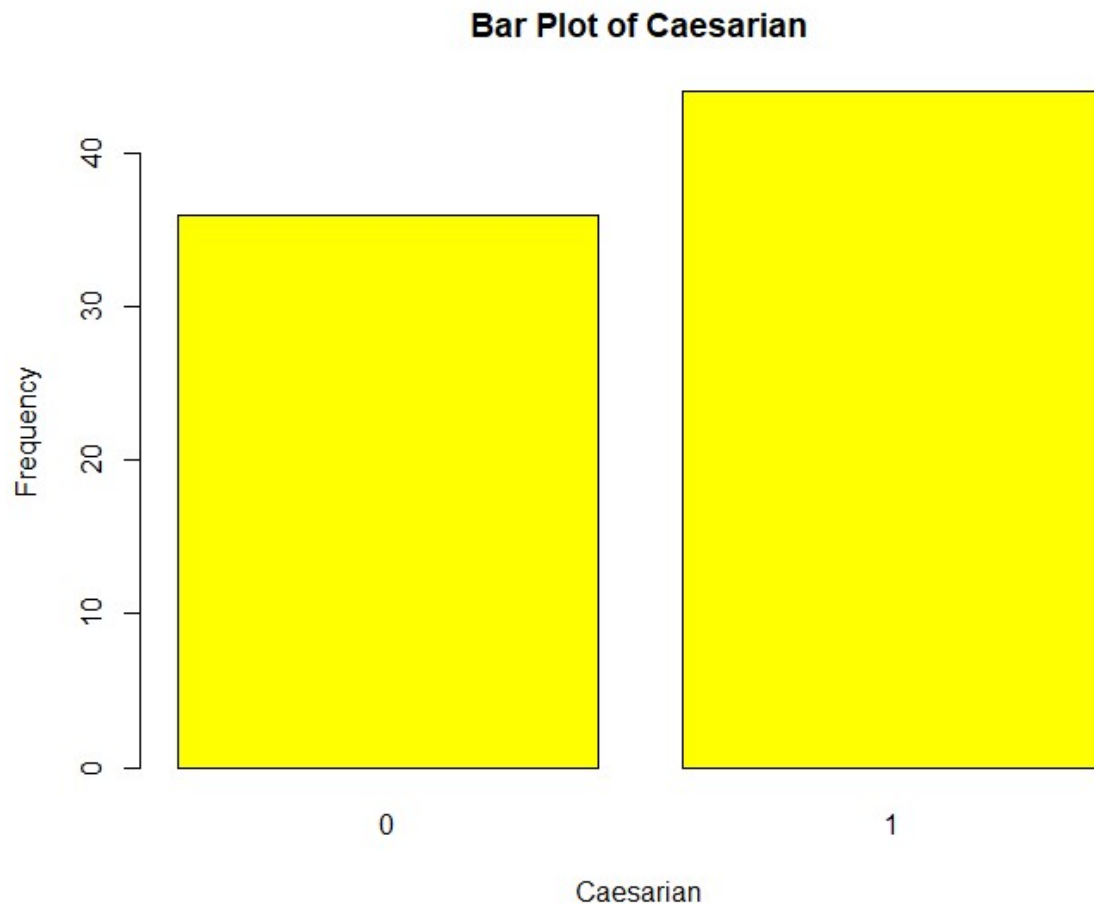
Bar Plot of Heart:

```
barplot(table(Dataset$Heart), main = "Bar Plot of Heart", xlab = "Heart", ylab = "Frequency", col = "lightgreen")
```



Bar Plot of Caesarian:

```
barplot(table(Dataset$Caesarian), main = "Bar Plot of Caesarian", xlab = "Caesarian", ylab = "Frequency", col = "yellow")
```



Conclusion: The "Maternal Health and Caesarian Section Outcomes" Dataset preparation and study with R programming enabled for effective management of missing information and provided key insights into the dataset's features. Using data preparation methods, the problem of missing values was handled using a variety of strategies, including imputation using the mean, median, and mode, as well as the deletion of rows with missing values. According to the study, it is possible to replace missing values with the mean, median, or mode since this provides a balanced adjustment with the least detrimental impact on the dataset's overall distribution. However, the nature of the data and its potential influence on subsequent research must be carefully considered. Furthermore, in cases when there is little to no missing data and deleting those rows has little to no effect on the dataset's representativeness, the alternative technique of removing rows with missing values may be suitable. To avoid any biases caused by the removal of specific observations, caution should be exercised. I've also presented univariate data exploration techniques using barplots. These measures have significantly enhanced the dataset's quality, opening the door for more complicated modelling and prediction tasks. It has been demonstrated how versatile R programming is in managing missing data and performing univariate analysis, emphasizing the validity and interpretability of outcomes. The combination of creative univariate examination and effective data preparation, particularly in the prediction of Maternal Health and Caesarean Section Outcomes, creates a solid foundation for further analysis and study.

References:

- [1] Josse, Julie, and François Husson. "Handling missing values in exploratory multivariate data analysis methods." *Journal de la société française de statistique* 153.2 (2012): 79-99.
- [2] Liew, A. W. C., Law, N. F., & Yan, H. (2011). Missing value imputation for gene expression data: computational techniques to recover missing data from available information. *Briefings in bioinformatics*, 12(5), 498-513.
- [3] Palanivinayagam, Ashokkumar, and Robertas Damaševičius. "Effective Handling of Missing Values in Datasets for Classification Using Machine Learning Methods." *Information* 14.2 (2023): 92.
- [4] Souza, J.P., Gülmezoglu, A.M., Lumbiganon, P., Laopaiboon, M., Carroli, G., Fawole, B., Ruyan, P. and WHO Global Survey on Maternal and Perinatal Health Research Group, 2010. Caesarean section without medical indications is associated with an increased risk of adverse short-term maternal outcomes: the 2004-2008 WHO Global Survey on Maternal and Perinatal Health. *BMC medicine*, 8, pp.1-10.
- [5] Rawal, Shivani, S. C. Gupta, and Shekhar Singh. "Predicting missing values in a dataset: challenges and approaches." *International Journal of Recent Research Aspects* 4.3 (2017): 34-38.
- [6] Kaiser, Jiří. "Dealing with Missing Values in Data." *Journal Of Systems Integration* (1804-2724) 5.1 (2014).
- [7] Shah, A., Fawole, B., M'imunya, J.M., Amokrane, F., Nafiou, I., Wolomby, J.J., Mugerwa, K., Neves, I., Nguti, R., Kublickas, M. and Mathai, M., 2009. Cesarean delivery outcomes from the WHO global survey on maternal and perinatal health in Africa. *International Journal of Gynecology & Obstetrics*, 107(3), pp.191-197.
- [8] Sessa, Jadran, and Dabeeruddin Syed. "Techniques to deal with missing data." 2016 5th international conference on electronic devices, systems and applications (ICEDSA). IEEE, 2016.
- [9] Doshi, Bhavik. "Handling Missing Values in Data Mining." *Data Cleaning and Preparation Term Paper* (2011).
- [10] Raju, K.B. and Rao, C.S., 2020. Classification of Caesarian Data Using Machine Learning Models. *EXECUTIVE EDITOR*, 11(03), p.3155.
- [11] Poolsawad, Nongnuch, et al. "Handling missing values in data mining-A case study of heart failure dataset." 2012 9th International Conference on Fuzzy Systems and Knowledge Discovery. IEEE, 2012.
- [12] Srivatsan, S. and Santhanam, T., APPLICATION OF ENSEMBLE CLASSIFIERS IN CAESARIAN SECTION PREDICTION.